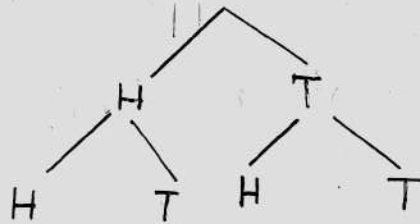


① Sample space: The sample space (S) of an experiment is a set, consisting of all of the possible experimental outcomes.

② Example: A usual six-sided dice has a sample space $S = \{1, 2, 3, 4, 5, 6\}$

③ If you toss a coin, the sample space $S = \{H, T\}$

④ What is the sample space when a coin is tossed two times? Or what is the sample space, if you toss two coins?

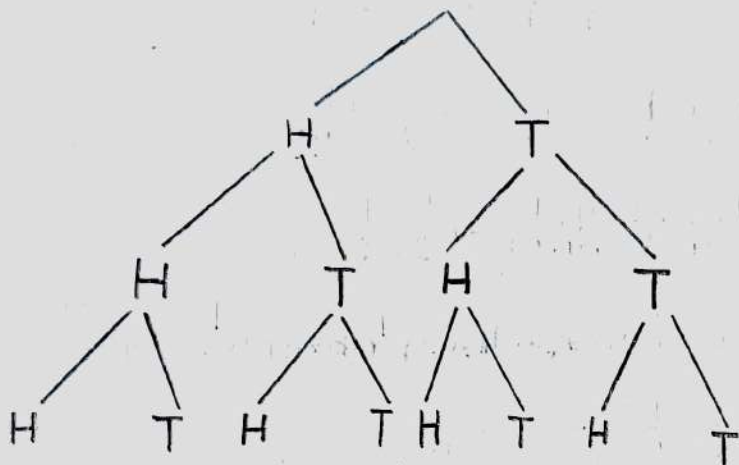


So, if I toss a coin two times, the possible outcomes are

$$S = \{HH, HT, TH, TT\}$$

Q (i) By using tree diagram write down the sample space when a coin is tossed three times?

Ans:



$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

(ii) What is the sample space for choosing a prime number less than 15 at random?

Ans: $S = \{2, 3, 5, 7, 11, 13\}$

Note: A prime number is a number greater than one, that is only divisible by one and itself.

(iii) If you throw two dice,

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$S = \{(1,1), (1,2), \dots \dots (6,6)\}$$

② Probability Values: Each outcome in the sample space has a probability value.

$$\text{Probability value} = \frac{\text{the number of favourable outcomes}}{\text{the total number of possible outcomes}}$$

For example, if you toss a coin, possible outcomes are $\{H, T\}$.

$$P(H) = \frac{1}{2}$$

$$P(T) = \frac{1}{2}$$

Properties of Probability Values:

(i) $0 \leq p \leq 1$

(ii) $\sum P = 1$

**** Note: If all the outcomes of a sample space have the same chance of occurrence then probability value

$$= \frac{1}{\text{the total numbers of possible outcomes}}$$

Example:

A fair die have six (n) outcomes and each of the six outcomes must have a probability of $\frac{1}{6}$,

i.e. $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

On a fair die, every numbers has an equal chance of being rolled. On a biased die, some numbers are more likely to be rolled than others. If a red die and a blue die are thrown,

with each of the 36 outcomes, let B be the event that at least one 6 is obtained on the two dice.

[Hint: look at the table]

So, the probability, $P(B) = \frac{11}{36}$

③ Events: An event is a subset of the sample space.

Ex: $S = \{1, 3, 4\}$ and $A = \{4\}$, since A is a subset of the sample space S . So, A is an event.

④ Complements of Events: The complement of an event A , is the event consisting of everything in the sample space S , that is not contained within the event A .

Ex: A usual six-sided die has a sample space $S = \{1, 2, 3, 4, 5, 6\}$, If Event $A = \{1, 3, 6\}$ then complement of event $A = \{2, 4, 5\}$.

⑤ Intersections of Events: contains common outcomes between two events.

$$\text{Ex: } A = \{1, 3, 4\}, B = \{4, 5\}$$

$$A \cap B = \{4\}$$

⑥ Mutually Exclusive Events: Two events A and B are said to be mutually exclusive if they have no outcomes in common.

$$\text{Ex: } A = \{1, 3, 4\}, B = \{5\}$$

Since event A and B don't have any common outcome, so they are Mutually Exclusive Events.

⑦ Unions of Events: consists of the outcomes that are contained within at least one of the events A and B.

$$\text{Ex: } A = \{1, 3, 4\}, B = \{4, 5\}$$

$$A \cup B = \{1, 3, 4, 5\}$$

① Q: An experiment has five outcomes, a, b, c, d, and e.

If $p(a) = 0.08$, $p(b) = 0.20$ and $p(c) = 0.33$, what are the possible values for the probability of outcome e. If outcome d and e are equal likely, what are their probable values.

Solution: We know,

$$\sum_{i=1}^n p_i = 1$$

$$\Rightarrow p(a) + p(b) + p(c) + p(d) + p(e) = 1$$

$$\Rightarrow 0.08 + 0.20 + 0.33 + p(d) + p(e) = 1$$

$$\Rightarrow 0.61 + p(d) + p(e) = 1$$

$$\Rightarrow p(d) + p(e) = 1 - 0.61$$

$$\therefore p(d) + p(e) = 0.39 \quad \text{--- (1)}$$

$$\text{Now, } p(d) = 0.39 - p(e)$$

$$\therefore 0 \leq p(e) \leq 0.39$$

$$\text{Hence, } p(e) = p(d)$$

$$\text{① becomes, } p(e) + p(e) = 0.39$$

$$\Rightarrow 2 \times p(e) = 0.39$$

$$\Rightarrow p(e) = 0.195$$

$$\therefore p(d) = 0.195$$

If d and e are equal likely,

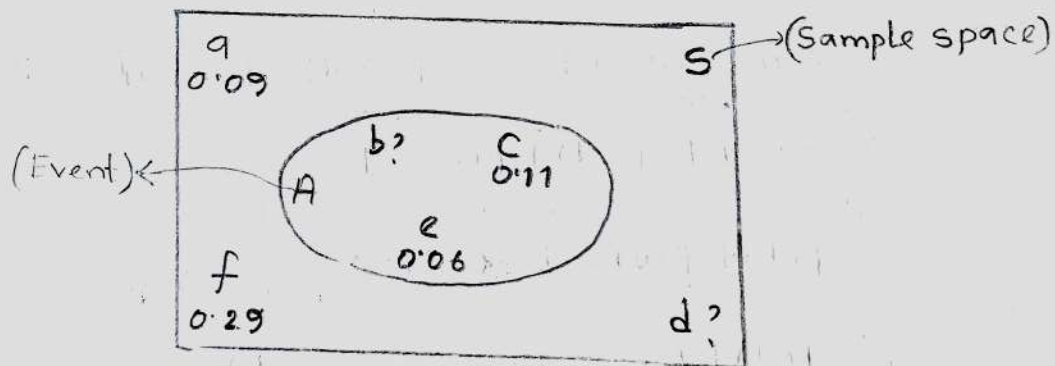
then probability = $\frac{2}{5}$. $\left[\frac{1}{5} + \frac{1}{5} = \frac{1+1}{5} = \frac{2}{5} \right]$

(Ans:)

② Q: Consider the sample space consist of a, b, c, d and e outcomes.

If $P(A) = 0.27$, calculate a) $P(b) = ?$, b) $P(A') = ?$

c) $P(d) = ?$



Solution: a) Given,

$$P(A) = 0.27$$

$$\Rightarrow P(b) + P(c) + P(e) = 0.27$$

$$\Rightarrow P(b) + 0.11 + 0.06 = 0.27$$

$$\Rightarrow P(b) = 0.27 - 0.17$$

$$\Rightarrow P(b) = 0.1$$

$$b) P(A') = P(a) + P(f) + P(d)$$

We know,

$$P(a) + P(b) + P(c) + P(d) + P(e) + P(f) = 1$$

$$\Rightarrow 0.09 + 0.1 + 0.11 + P(d) + 0.06 + 0.29 = 1$$

$$\Rightarrow P(d) + 0.65 = 1$$

$$\Rightarrow P(d) = 1 - 0.65$$

$$= 0.35$$

$$\therefore P(A') = 0.09 + 0.29 + 0.35$$

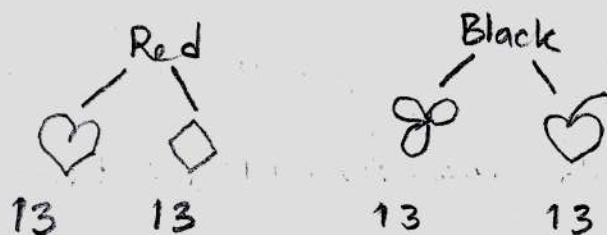
$$= 0.73$$

c) From (b), we can see $P(d) = 0.35$

(Ans.)

- ③ Q: If a card is chosen at random from a pack of cards, what is the probability that the card is from one of the two black sets.

Solution: 52 cards

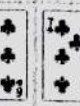
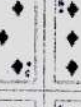
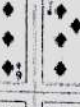
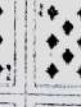
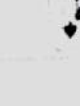


$P(\text{card is from one of the two black sets})$

$$= \frac{\text{Number of favourable outcomes}}{\text{Total no. of possible outcomes}}$$

$$= \frac{26}{52}$$

Example set of 52 poker playing cards

Suit	Ace	2	3	4	5	6	7	8	9	10	Jack	Queen	King
Clubs													
Diamonds													
Hearts													
Spades													

④ Q: Three types of batteries are being tested type I, type II, and type III. The outcome (I, II, III) denotes that the battery of type I fails first, type II fails next and type III fails last. So, type III last the longest. The probability of the six outcomes are given below.

$$P(I, II, III) = 0.11, \quad P(I, III, II) = 0.07, \quad P(II, I, III) = 0.24$$

$$P(II, III, I) = 0.39, \quad P(III, I, II) = 0.16, \quad P(III, II, I) = 0.03$$

What is the probⁿ that

a) type I battery last longest?

b) type I battery doesn't last longest?

c) type I battery last longer than type II battery?

Solution:

$$(a) \quad P(II, III, I) + P(III, II, I) = 0.39 + 0.03 \\ = 0.42$$

$$(b) \quad P(I, II, III) + P(I, III, II) + P(II, I, III) + P(III, I, II) \\ = 0.11 + 0.07 + 0.24 + 0.16 \\ = 0.58$$

$$(c) P(II, I, III) + P(II, III, I) + P(III, II, I)$$

$$= 0.24 + 0.39 + 0.03$$

$$= 0.66$$

⑤ Q: A card is chosen from a pack of cards.

Q Are the events that a card from one of the two red sets is chosen and a card from one of the two black sets is chosen mutually exclusive?

Ans: Yes!

Q What about the events that an ace is chosen and a heart is chosen?

Ans: No. They are not mutually exclusive.

⑥ Q: If $P(A) = 0.4$ and $P(A \cap B) = 0.3$, what are the possible values of $P(B)$.

Solⁿ: We know,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Now,

$$P(A) + P(B) - P(A \cap B) \leq 1$$

$$\Rightarrow P(B) \leq 1 - P(A) + P(A \cap B)$$

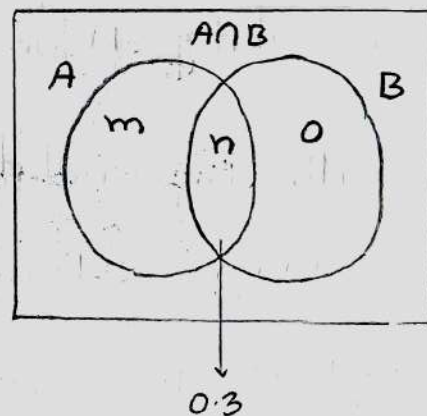
$$\Rightarrow P(B) \leq 1 - 0.4 + 0.3$$

$$\Rightarrow P(B) \leq 0.9$$

Also, $P(B) \geq P(A \cap B)$

$$\therefore 0.3 \leq P(B) \leq 0.9$$

(Ans.)



$$P(A) = P(m) + P(n)$$

$$P(B) = P(o) + P(n)$$

$$P(A \cap B) = P(n)$$

$$\text{i.e. } P(B) \geq P(A \cap B)$$

Q Conditional probability:

Conditional probability is the probability of an event given that another event has already occurred.

Conditional probability is denoted by

$$P(A|B) = P(A \cap B) / P(B)$$

Example: A manager supervises the operation of three power plants, plant X, plant Y, and plant Z. At any given time, each of the three plants can be classified as either generating electricity (1) or not generating electricity (0).

With the notation (0,1,0) used to represent the situation where plant Y is generating electricity but plants X and Z are both not generating electricity [Let A be the event that plant X is not generating electricity and B be the event that at least two out of the three plants are generating electricity.]

5

$(0, 0, 0)$ 0.07 $(0, 0, 1)$ 0.04 $(0, 1, 0)$ 0.03 $(0, 1, 1)$ 0.18	$(1, 0, 0)$ 0.16 $(1, 0, 1)$ 0.18 $(1, 1, 0)$ 0.21 $(1, 1, 1)$ 0.13
--	--

(i) If event B is known what is the probability of happening the event A?

(ii) What is the probability of happening the event A?

Solution:

$$\begin{aligned}
 (i) \quad P(B) &= P(1, 0, 1) + P(1, 1, 0) + P(1, 1, 1) + P(0, 1, 1) \\
 &= 0.18 + 0.21 + 0.13 + 0.18 \\
 &= 0.70
 \end{aligned}$$

$$\begin{aligned}
 P(A \cap B) &= P(0, 1, 1) \\
 &= 0.18
 \end{aligned}$$

$$\begin{aligned} \therefore P(A|B) &= P(A \cap B) / P(B) \\ &\stackrel{\text{(If known)}}{=} 0.18 / 0.70 \\ &= 0.257 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(A) &= P(0, 0, 0) + P(0, 0, 1) + P(0, 1, 0) + P(0, 1, 1) \\ &= 0.07 + 0.04 + 0.03 + 0.18 \\ &= 0.32 \end{aligned}$$

Q A card is drawn at random from a pack of cards. Calculate:

$$\textcircled{a} P(\text{King | card from red suit}) = ?$$

$$A = \{K \text{ clubs, } K \text{ diamonds, } K \text{ hearts, } K \text{ spades}\}$$

$$B = \{13 \text{ diamonds, } 13 \text{ hearts}\}$$

$$A \cap B = \{K \text{ diamonds, } K \text{ hearts}\}$$

$$\therefore P(A \cap B) = \frac{2}{52}$$

$$|B| = (13 + 13) = 26$$

$$\therefore P(B) = \frac{26}{52}$$

$$P(A|B) = P(A \cap B) / P(B)$$

$$= \frac{\left(\frac{2}{52}\right)}{\left(\frac{26}{52}\right)}$$

$$= \frac{1}{13}$$

(Ans:)

⑥ $P(\text{king} | \text{red picture card}) = ?$

$$A = \{K \text{ clubs, } K \text{ diamonds, } K \text{ hearts, } K \text{ spades}\}$$

$$B = \{J \text{ diamonds, } J \text{ hearts, } Q \text{ diamonds, } Q \text{ hearts, } K \text{ diamonds, } K \text{ hearts}\}$$

$$A \cap B = \{K \text{ diamonds, } K \text{ hearts}\}$$

$$\therefore P(A \cap B) = \frac{2}{52}$$

$$|B| = 6$$

$$\therefore P(B) = \frac{6}{52}$$

$$\therefore P(A|B) = P(A \cap B) / P(B)$$

$$= \frac{\left(\frac{2}{52}\right)}{\left(\frac{6}{52}\right)}$$

$$= \frac{1}{3}$$

(Ans:)

(c) $P(\text{heart} \mid \text{card from black suit}) = ?$

$$A = \{13 \text{ hearts}\}$$

$$B = \{13 \text{ clubs}, 13 \text{ spades}\}$$

$$\therefore A \cap B = \{\}$$

$$\therefore P(A|B) = P(A \cap B) / P(B)$$

$$= \frac{0}{\left(\frac{26}{52}\right)}$$

$$= 0$$

(Ans.)

Question: A car repair is either on time or late and either satisfactory or unsatisfactory. If a repair is made on time, then there is a probability of 0.85 that it is satisfactory. There is a probability of 0.77 that a repair will be made on time. What is the probability that a repair is made on time and is satisfactory?

Solution:

Let,

If a car repairment is satisfactory, then it is denoted by S ,

If a car repairment is unsatisfactory, then it is denoted by S' ,

If a car repairment is on time then it is denoted by T ,

If a car repairment is on late then it is denoted by T'

Here,

$$P(S|T) = 0.85$$

$$P(T) = 0.77$$

$$P(S \cap T) = ?$$

we know,

$$P(S|T) = P(S \cap T) / P(T)$$

$$\Rightarrow P(S \cap T) = P(S|T) * P(T)$$

$$= 0.85 \times 0.77$$

$$= 0.6545$$

(Ans.)

Question: Three types of batteries are being tested, type I, type II, and type III. The outcome (I, II, III) denotes that the battery of type I fails first, the battery of type II next, and the battery of type III lasts the longest.

(I, II, III) 0.11	(I, III, II) 0.07
(II, I, III) 0.24	(II, III, I) 0.39
(III, I, II) 0.16	(III, II, I) 0.03

- (a) A type I battery lasts longest conditional on a type II battery failing first.
- (b) A type I battery lasts longest conditional on a type II battery lasting the longest.

Solution: Let A denotes type I battery lasts longest
and B denotes type II battery failing first.

①

$$A = \{(II, III, I), (III, II, I)\}$$

$$B = \{(II, I, III), (II, III, I)\}$$

$$A \cap B = \{(II, III, I)\}$$

$$\therefore P(A \cap B) = 0.39$$

$$P(B) = (0.24) + (0.39)$$

$$= 0.63$$

$$\therefore P(A|B) = P(A \cap B) / P(B)$$

$$= 0.39 / 0.63$$

$$= 0.62$$

(Ans.)

②

$$A = \{(II, III, I), (III, II, I)\}$$

$$B = \{(I, III, II), (III, I, II)\}$$

$$\therefore A \cap B = \{\}$$

$$\therefore P(A \cap B) = 0$$

$$\therefore P(A|B) = P(A \cap B) / P(B)$$

$$= 0 / P(B)$$

$$= 0$$

(Ans.)

© A type I battery lasts longest conditional on a type II battery not failing first.

$$A = \{(II, III, I), (I, II, I)\}$$

$$B = \{(I, II, III), (I, III, II), (III, I, II), (III, II, I)\}$$

$$\therefore A \cap B = \{(III, II, I)\}$$

$$P(A \cap B) = 0.03$$

$$P(B) = (0.11) + (0.07) + (0.16) + (0.03)$$

$$= 0.37$$

$$\therefore P(A|B) = P(A \cap B) / P(B)$$

$$= 0.03 / 0.37$$

$$= 0.08$$

(Ans.)

⊕ Random Variable:

A random variable is obtained by assigning a numerical value to each outcome of a particular experiment.

Random Variable	Possible Values	Random Events
$X =$	0	(H)
	1	(T)

H T

$$S = \{H, T\}$$

$$X = \{0, 1\}$$

↪ number of heads

So,

we have experiment (such as tossing a coin)

We give values to each event

The set of values is a Random Variable

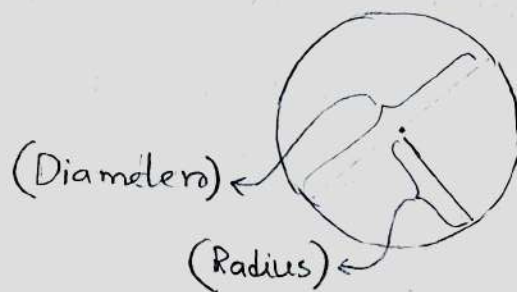
⊕ Types of a Random Variable:

Random Variables can be either Discrete or Continuous. Discrete random variable can only take certain values (such as 0, 1). Continuous random variable can take any value within a range.

Example: A company manufactures metal cylinders that are used in the construction of a particular type of engine. The company discovers that the cylinders it manufactures can have a diameter anywhere between 49.5 mm and 50.5 mm. Suppose that the random variable X is the diameter of a randomly chosen cylinder manufactured by the company. Since this random variable can take any value between 49.5 and 50.5, it is a continuous random variable.

Discrete RV: $X: 0, 1$

Continuous RV: $49.5 \leq X \leq 50.5$



Q. Probability mass function (PMF):

PMF is a function of a discrete random variable that gives the probability that a discrete random variable is exactly equal to a value. Note that by definition the PMF is a probability measures, so it satisfies all properties of a probability. i.e The properties of PMF:

$$(i) 0 \leq P_i \leq 1$$

$$(ii) \sum P_i = 1$$

Example: I toss a coin twice. Let X be the number of heads. Find the PMF of X .

Solution: Here, the random variable X be the number of heads. If I toss a coin twice, the possible outcomes are HH, HT, TH, TT.

$$\therefore S = \{HH, HT, TH, TT\}$$

So, the value of the $X = \{0, 1, 2\}$

$$\text{Here, } P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}$$

PMF is given by,

$$P(0) = P(T) = \frac{1}{4}$$

$$P(1) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P(2) = P(HH) = \frac{1}{4}$$

(Ans.)

Example: I toss a coin. Let X be the number of heads. Find the PMF of X .

Solution:

Here,

the random variable X be the number of heads. If I toss a coin, the possible outcomes are,

$$S = \{H, T\}$$

So, the value of the $X = \{0, 1\}$

$$\therefore P(H) = \frac{1}{2}$$

$$P(T) = \frac{1}{2}$$

PMF is given by,

$$P(0) = P(T) = \frac{1}{2}$$

$$P(1) = P(H) = \frac{1}{2}$$

(Ans:)

Example: Given,

$X = \{1, 2, 3, 4, 5, 6\}$ (the score shown on top face)

$$S = \{1, 2, 3, 4, 5, 6\}$$

Hence, $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

$\therefore$ PMF is given by,

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

(Ans:)

Example: If you throw a dice twice. Let X be the number of 6 comes on the two dice. Find the probability mass function of X .

Solution: Hence,

the number of total possible outcomes is 36.

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Here, the value of $X : \{0, 1, 2\}$

$\therefore$ PMF is given by,

$$P(0) = \frac{25}{36} \text{ (the probability that no 6 comes on two dice)}$$

$$P(1) = \frac{10}{36} \text{ (the probability that one 6 comes on two dice)}$$

$$P(2) = \frac{1}{36} \text{ (the probability that two 6 comes on two dice)}$$

Verify: $P(0) + P(1) + P(2)$

$$= \frac{25}{36} + \frac{10}{36} + \frac{1}{36}$$

$$= 1$$

(Ans.)

Q Cumulative distribution function (CDF):

CDF is a function that starts at a smallest value of x (numerical values of the random variable X) and increases to the largest values of x .

Example: I toss a coin twice. Let X be the numbers of heads. Find the CDF of X .

Solution: The random variable X be the numbers of heads. If I toss a coin twice, then the numbers of possible outcomes are,

$$S = \{HH, HT, TH, TT\}$$

So, the value of the $X = \{0, 1, 2\}$

$\therefore$ PMF is given by,

$$P(0) = P(TT) = \frac{1}{4}$$

$$P(1) = P(HT) + P(TH) = \left(\frac{1}{4}\right) + \left(\frac{1}{4}\right) = \frac{1}{2}$$

$$P(2) = P(HH) = \frac{1}{4}$$

CDF is given by,

$$F(0) = P(0) = \frac{1}{4} \text{ [probability that no head occurs]}$$

$$\begin{aligned} F(1) &= P(0) + P(1) = \frac{1}{4} + \frac{1}{2} \text{ [probability that not more} \\ &= \frac{3}{4} \text{ than 1 head occurs,]} \end{aligned}$$

$$\begin{aligned}
 F(2) &= P(0) + P(1) + P(2) \quad [\text{probability that no more} \\
 &= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} \quad \text{than two heads occur}] \\
 &= 1
 \end{aligned}$$

(Ans:)

Example: A manager supervises the operations of three power plants, plant X, plant Y, and plant Z. At any given time, each of the three plants can be classified as either generating electricity (1) or being idle (0). with the notation $(0, 1, 0)$ used to represent the situation where plant Y is generating electricity but plant X and Z are not generating electricity.

1) Calculate the probability mass function.

2) Draw PMF

3) Calculate CDF

4) Draw CDF

$(0, 0, 0)$ 0.07	$(1, 0, 0)$ 0.16
$(0, 0, 1)$ 0.04	$(1, 0, 1)$ 0.18
$(0, 1, 0)$ 0.03	$(1, 1, 0)$ 0.21
$(0, 1, 1)$ 0.18	$(1, 1, 1)$ 0.13

Solution: Hence, random variable X be the numbers of plants generating electricity.

$$\therefore X = \{0, 1, 2, 3\}$$

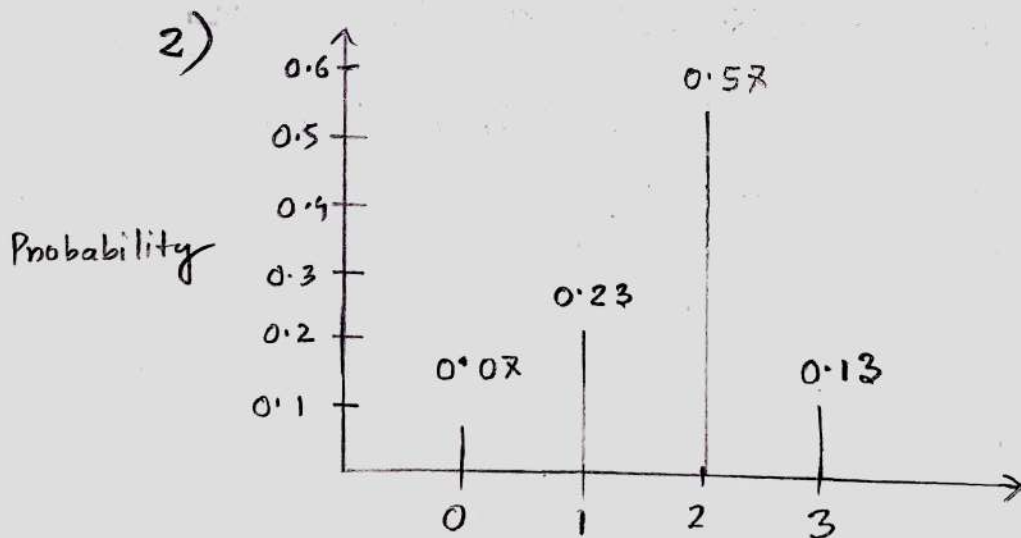
1) $P(0) = 0.07$ [probability that no plants are generating electricity]

$$P(1) = (0.04) + (0.03) + (0.16) \text{ [probability that one plant is generating electricity]}$$
$$= 0.23$$

$$P(2) = (0.18) + (0.18) + (0.21) \text{ [probability that two plants are generating electricity]}$$
$$= 0.57$$

$$P(3) = 0.13 \text{ [probability that three plants are generating electricity]}$$

(Ans.)



Number of power plants generating electricity

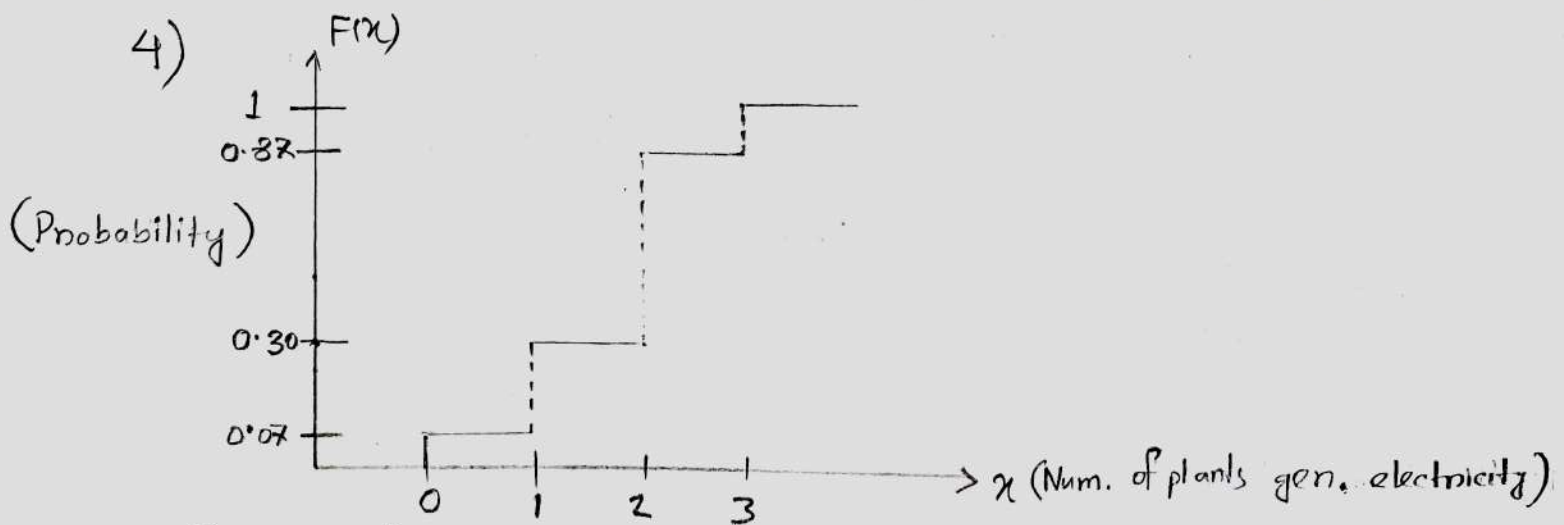
3) CDF is given by,

$$F(0) = P(0) = 0.07 \quad [\text{probability that no plants are generating electricity}]$$

$$\begin{aligned} F(1) &= P(0) + P(1) \quad [\text{probability that no more} \\ &= 0.07 + 0.23 \quad \text{than one plant is generating} \\ &= 0.30 \quad \text{electricity}] \end{aligned}$$

$$\begin{aligned} F(2) &= P(0) + P(1) + P(2) \quad [\text{probability that no more than} \\ &= 0.30 + 0.57 \quad \text{two plants are generating} \\ &= 0.87 \quad \text{electricity}] \end{aligned}$$

$$\begin{aligned} F(3) &= P(0) + P(1) + P(2) + P(3) \quad [\text{probability that no more} \\ &= 0.87 + 0.13 \quad \text{than three plants are} \\ &= 1 \quad \text{generating electricity}] \end{aligned}$$



Graph of the cumulative distribution function.

Classwork-1: An office has four copying machine, and the random variable X measures how many of them are in use at a particular moment in time.

Suppose that

$$P(X=0) = 0.08, P(X=1) = 0.11, P(X=2) = 0.27, \text{ and } P(X=3) = 0.33$$

(a) What is $P(X=4)$?

(b) Draw a line graph of the probability mass function.

(c) Construct and plot the cumulative distribution function.

Solution: (a) We know,

$$X = \{0, 1, 2, 3, 4\}$$

(num. of active copying machine)

$$\sum p_i = 1$$

$$\therefore P(0) + P(1) + P(2) + P(3) + P(4) = 1$$

$$\Rightarrow 0.08 + 0.11 + 0.27 + 0.33 + P(4) = 1$$

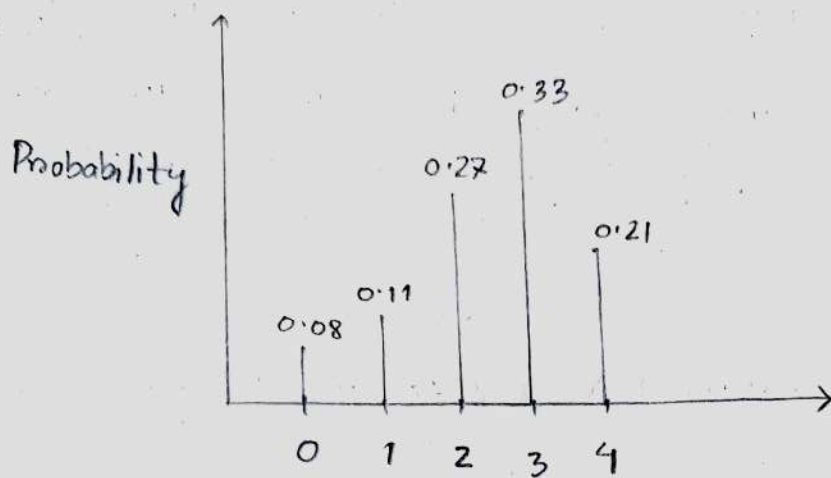
$$\Rightarrow P(4) = 1 - 0.79$$

$$= 0.21$$

$$\therefore P(X=4) = 0.21$$

(Ans.)

(b)



Number of active copying machines.

(Ans.)

(c) Cumulative distribution function is constructed below.

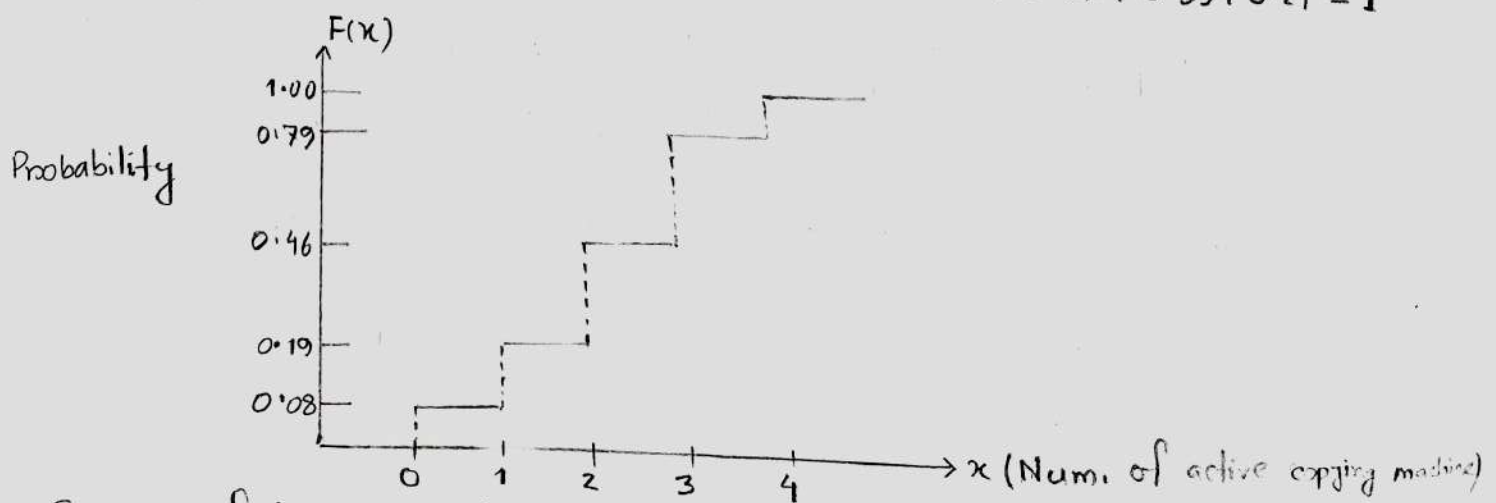
$$F(0) = P(0) = 0.08$$

$$F(1) = P(0) + P(1) = 0.08 + 0.11 = 0.19$$

$$F(2) = P(0) + P(1) + P(2) = 0.08 + 0.11 + 0.27 = 0.46$$

$$F(3) = P(0) + P(1) + P(2) + P(3) = 0.08 + 0.11 + 0.27 + 0.33 = 0.79$$

$$F(4) = P(0) + P(1) + P(2) + P(3) + P(4) = 0.08 + 0.11 + 0.27 + 0.33 + 0.21 = 1$$



Graph of the cumulative distribution function (Ans.)

Q Probability Density Function:

Probability density function is a function of continuous random variable.

The probability that the random variable lies between two values a and b is obtained by integrating the probability density function between these two values. A valid probability density function $f(x)$ must integrate to one over the whole sample space, so that the total probability is equal to 1.

Q Question: Suppose that the diameter of a metal cylinder has a density function $f(x) = 1.5 - 6(x - 50.0)^2$ for $49.5 \leq x \leq 50.5$.

- (i) Show that total area under the probability density function = 1 or Prove that this is a valid PDF.
- (ii) Calculate the probability that a metal cylinder has a diameter between 49.8 and 50.1 mm.
- (iii) Find cumulative distribution function.

Solution:

$$(i) \int_{49.5}^{50.5} [1.5 - 6(x-50.0)^2] dx$$

$$= \left[1.5x - \frac{6(x-50)^3}{3} \right]_{49.5}^{50.5}$$

$$= \left[1.5 \times (50.5) - \frac{6(50.5-50)^3}{3} \right] - \left[1.5 \times (49.5) - \frac{6(49.5-50)^3}{3} \right]$$

$$= 75.5 - 74.5$$

$$= 1$$

That is 100% of the cylinders will have diameters within these limits.

Since over the whole sample space, the integration of $f(x) = 1$, so this is a valid PDF.

$$(ii) \int_{49.8}^{50.1} [1.5 - 6(x-50.0)^2] dx$$

$$= \left[1.5x - \frac{6(x-50)^3}{3} \right]_{49.8}^{50.1}$$

$$= \left[1.5 \times (50.1) - \frac{6(50.1-50)^3}{3} \right] - \left[1.5 \times (49.8) - \frac{6(49.8-50)^3}{3} \right]$$

$$= 75.148 - 74.716$$

$$= 0.432 \approx 0.43$$

Ans,

$$\int c dx = cx$$

$$\int (x-a)^n dx = \frac{(x-a)^{n+1}}{n+1}$$

That is, 43% of the cylinders will have diameters within these limits.

$$(iii) F(x) = \int_{49.5}^x [1.5 - 6(x-50.0)^2] dx$$

$$= \left[1.5x - \frac{6(x-50)^3}{3} \right]_{49.5}^x$$

$$= \left[1.5x - \frac{6(x-50)^3}{3} \right] - \left[1.5 \times (49.5) - \frac{6(49.5-50)^3}{3} \right]$$

$$= [1.5x - 2(x-50)^3] - 74.5$$

▣ Question: $f(x) = A(0.5 - (x - 0.25)^2)$ for $0.125 \leq x \leq 0.5$

(a) Find the value of A .

(b) Construct the cumulative distribution function.

(c) What is the probability that the paint thickness at a particular point is less than 0.2 mm?

P.T.O.

Solution:

$$(a) \int_{0.125}^{0.5} A(0.5 - (x - 0.25)^2) dx = 1$$

$$\Rightarrow A \left[0.5x - \frac{(x - 0.25)^3}{3} \right]_{0.125}^{0.5} = 1$$

$$\Rightarrow A \left[\left[0.5 \times (0.5) - \frac{(0.5 - 0.25)^3}{3} \right] - \left[0.5 \times (0.125) - \frac{(0.125 - 0.25)^3}{3} \right] \right] = 1$$

$$\Rightarrow A \left[\frac{47}{192} - \frac{97}{1536} \right] = 1$$

$$\Rightarrow A \frac{93}{512} = 1$$

$$\Rightarrow A = \frac{512}{93}$$

$$(b) F(x) = \int_{0.125}^x \frac{512}{93} (0.5 - (x - 0.25)^2) dx$$

$$= \frac{512}{93} \left[0.5x - \frac{(x - 0.25)^3}{3} \right]_{0.125}^x$$

$$= \frac{512}{93} \left[\left[0.5x - \frac{(x - 0.25)^3}{3} \right] - \left[0.5 \times (0.125) - \frac{(0.125 - 0.25)^3}{3} \right] \right]$$

$$= \frac{512}{93} \left[\left[0.5x - \frac{(x - 0.25)^3}{3} \right] - \frac{97}{1536} \right]$$

$$(c) \int_{0.125}^{0.2} \frac{512}{93} (0.5 - (x - 0.25)^2) dx$$

$$= \frac{512}{93} \left[\left[0.5 \times (0.2) - \frac{(0.2 - 0.25)^3}{3} \right] - \frac{97}{1536} \right]$$

$$= 0.203$$

▣ Expectation of a Random Variable;

Expectation of a Random Variable is denoted by $E(X)$ and represents an 'average' value of the random variable.

For discrete random variable, $E(X) = \sum x p(x)$

For continuous random variable, $E(X) = \int_L^U x f(x) dx$

1 Example: A manager supervises the operation of three power plants, plant X, plant Y, and plant Z. At any given time, each of the three plants can be classified as either generating electricity (1) or being idle (0). With notation (0, 1, 0) used to represent the situation where plant Y is generating electricity but plants X and Z are not generating electricity. Here, random variable X be the numbers of plants generating electricity.

(1) Calculate the probability mass function.

(2) Draw PMF

(3) Draw CDF

(4) Draw CDF

* (5) Calculate the expected power plant generating electricity.

(0,0,0)	(1,0,0)
0.07	0.16
(0,0,1)	(1,0,1)
0.04	0.18
(0,1,0)	(1,1,0)
0.03	0.21
(0,1,1)	(1,1,1)
0.18	0.13

Solution: (5) Here, $X = \{0, 1, 2, 3\}$

$$P(0) = 0.07$$

$$P(1) = (0.16 + 0.04 + 0.03) = 0.23$$

$$P(2) = (0.18 + 0.21 + 0.18) = 0.57$$

$$P(3) = 0.13$$

$$\therefore E(X) = \sum x P(x)$$

$$= (0 \times 0.07) + (1 \times 0.23) + (2 \times 0.57) + (3 \times 0.13)$$

$$= 1.76$$

(Ans:)

2 Example: Suppose that the diameters of a metal cylinders has a probability density function

$$f(x) = 1.5 - 6(x - 50.0)^2 \quad \text{for } 49.5 \leq x \leq 50.5$$

- (i) Show that total area under the probability density function = 1 or Prove that this is a valid PDF.
- (ii) Calculate the probability that a metal cylinders has a diameters between 49.8mm and 50.1 mm.
- (iii) Find cumulative distribution function.
- * (iv) Calculate the expected diameters of a metal cylinders.

Solution: We know,

$$\begin{aligned} E(x) &= \int_L^U x f(x) dx \\ &= \int_{49.5}^{50.5} x (1.5 - 6(x - 50.0)^2) dx \\ &= \int_{49.5}^{50.5} x (1.5 - 6(x^2 - 100x + 2500)) dx \\ &= \int_{49.5}^{50.5} (1.5x - 6x^3 + 600x^2 - 15000x) dx \end{aligned}$$

$$= \left[1.5 \frac{x^2}{2} - 6 \frac{x^4}{4} + 600 \frac{x^3}{3} - 15000 \frac{x^2}{2} \right]_{49.5}^{50.5}$$

$$= (-3123099.906) - (3123149.906)$$

$$= -5000025$$

$$\approx 50$$

(Ans.)

▣ Variance: Variance measures how far a set of numbers are spread out from their average value.

For discrete random variable,

$$\begin{aligned} V(x) &= E(x^2) - (E(x))^2 \\ &= \sum x^2 p(x) - \left(\sum x p(x) \right)^2 \end{aligned}$$

For continuous random variable,

$$\begin{aligned} V(x) &= E(x^2) - (E(x))^2 \\ &= \int_L^U x^2 f(x) dx - \left[\int_L^U x f(x) dx \right]^2 \end{aligned}$$

Standard deviation: Standard deviation is the square root of the variance.

$$SD(x) = \sqrt{V(x)}$$

Example: Continuation from example 1 ...

* (6) Calculate variance and standard deviation.

Solution: Here, $X = \{0, 1, 2, 3\}$

So, $P(0) = 0.07$, $P(1) = 0.23$, $P(2) = 0.57$, $P(3) = 0.13$

$$\begin{aligned} E(X) &= \sum X P(X) = (0 \times 0.07) + (1 \times 0.23) + (2 \times 0.57) + (3 \times 0.13) \\ &= 1.76 \end{aligned}$$

$$\begin{aligned} E(X^2) &= \sum X^2 P(X) = (0^2 \times 0.07) + (1^2 \times 0.23) + (2^2 \times 0.57) + (3^2 \times 0.13) \\ &= 3.68 \end{aligned}$$

$$\therefore V(X) = E(X^2) - (E(X))^2$$

$$= 3.68 - (1.76)^2$$

$$= 0.5824$$

$$SD(X) = \sqrt{V(X)}$$

$$= \sqrt{0.5824}$$

$$= 0.7632$$

(Ans.)

Example: Continuation from example [2] ...

* (v) Calculate variance and standard deviation.

Solution: We know,

$$V(x) = E(x^2) - (E(x))^2$$

$$E(x) = 50 \text{ [from previous solution]}$$

$$E(x^2) = \int_L^U x^2 f(x) dx$$

$$= \int_{49.5}^{50.5} x^2 (1.5 - 6(x-50)^2) dx$$

$$= \int_{49.5}^{50.5} x^2 (1.5 - 6(x^2 - 100x + 2500)) dx$$

$$= \int_{49.5}^{50.5} (1.5x^2 - 6x^4 + 600x^3 - 2500x^2) dx$$

$$= \left[1.5 \frac{x^3}{3} - 6 \frac{x^5}{5} + 600 \frac{x^4}{4} - 2500 \frac{x^3}{3} \right]_{49.5}^{50.5}$$

$$= -62436241 + 62438741$$

$$= 2500$$

$$\therefore V(x) = 2500 - 50^2 = 0$$

$$SD(x) = \sqrt{V(x)} = 0$$

(Ans.)

Classwork-2: A random variable X takes values between 4 and 6 with a probability density function

$$f(x) = \frac{1}{x \ln(1.5)} \quad \text{for } 4 \leq x \leq 6$$

(i) Check the total area under the probability density function is equal to 1.

(ii) What is $P(4.5 \leq x \leq 5.5)$?

(iii) Calculate the cumulative distribution function.

Solution:

$$(i) \int_4^6 \frac{1}{x \ln(1.5)} dx$$

$$= \left[\frac{1}{\ln(1.5)} \ln x \right]_4^6$$

$$= \left[\frac{\ln 6}{\ln(1.5)} - \frac{\ln 4}{\ln(1.5)} \right]$$

$$= \frac{\ln 6 - \ln 4}{\ln(1.5)}$$

$$= \frac{\ln \frac{6}{4}}{\ln(1.5)}$$

$$= \frac{\ln(1.5)}{\ln(1.5)}$$

$$= 1$$

(checked)

$$(ii) \int_{4.5}^{5.5} \frac{1}{x \ln(1.5)} dx$$

$$= \left[\frac{1}{\ln(1.5)} \ln x \right]_{4.5}^{5.5}$$

$$= \left[\frac{\ln(5.5)}{\ln(1.5)} - \frac{\ln(4.5)}{\ln(1.5)} \right]$$

$$= \frac{\ln(5.5) - \ln(4.5)}{\ln(1.5)}$$

$$= \frac{\ln \frac{5.5}{4.5}}{\ln(1.5)}$$

$$= 0.49$$

(Ans.)

$$\begin{aligned}
 \text{(iii)} \quad & \int_4^x \frac{1}{x \ln(1.5)} dx \\
 &= \left[\frac{1}{\ln(1.5)} \ln x \right]_4^x \\
 &= \left[\frac{\ln x}{\ln(1.5)} - \frac{\ln 4}{\ln(1.5)} \right] \\
 &= \frac{\ln x - \ln 4}{\ln(1.5)}
 \end{aligned}$$

(Ans:)

Classwork-3: Suppose that the random variable x measures the thickness of the paint in millimeters at a randomly chosen point on a randomly chosen car panel, and that ' x ' takes values between 0.125mm and 0.5mm with a probability density function of

$$f(x) = \frac{512}{93} (0.5 - (x - 0.25)^2) \quad \text{for } 0.125 \leq x \leq 0.5$$

- (i) what is the expected or average paint thickness?
- (ii) what is the variance and standard deviation?

Solution:

(i) For expected on average thickness

$$E(x) = \int_L^U x f(x) dx$$

$$= \int_{0.125}^{0.5} x \frac{512}{93} [0.5 - (x - 0.25)^2] dx$$

$$= \frac{512}{93} \int_{0.125}^{0.5} [0.5x - x(x^2 - 2 \times x \times 0.25 + 0.25^2)] dx$$

$$= \frac{512}{93} \int_{0.125}^{0.5} [0.5x - x^3 + 0.5x^2 - 0.0625x] dx$$

$$= \frac{512}{93} \left[0.5 \frac{x^2}{2} - \frac{x^4}{4} + 0.5 \frac{x^3}{3} - 0.0625 \frac{x^2}{2} \right]_{0.125}^{0.5}$$

$$= \frac{512}{93} \left[\frac{23}{384} - \frac{131}{49152} \right]$$

$$= \frac{512}{93} \times \frac{921}{16384}$$

$$= \frac{307}{992}$$

$$= 0.309$$

(Ans.)

(ii) For variance and standard deviation

$$V(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \int_L^U x^2 f(x) dx$$

$$= \int_{0.125}^{0.5} x^2 \frac{512}{93} [0.5 - (x - 0.25)^2] dx$$

$$= \frac{512}{93} \int_{0.125}^{0.5} [0.5x^2 - x^2(x^2 - 0.5x + 0.0625)] dx$$

$$= \frac{512}{93} \int_{0.125}^{0.5} [0.5x^2 - x^4 + 0.5x^3 - 0.0625x^2] dx$$

$$= \frac{512}{93} \left[0.5 \frac{x^3}{3} - \frac{x^5}{5} + 0.5 \frac{x^4}{4} - 0.0625 \frac{x^3}{3} \right]_{0.125}^{0.5}$$

$$= \frac{512}{93} \left[\frac{19}{960} - \frac{19}{61440} \right]$$

$$= \frac{512}{93} \times \frac{399}{20480}$$

$$= \frac{133}{1240} = 0.107$$

$$\therefore V(x) = E(x^2) - (E(x))^2$$

$$= 0.107 - (0.309)^2 = 0.011519$$

$$SD = \sqrt{V(x)} = 0.107$$

(Ans.)

Quantiles: Quantiles are the three cut points that will divide a dataset into four equal-sized groups.

1) 1st quantile / lower quantile

2) 2nd quantile / Median

3) 3rd quantile / Upper quantile

(1) 1st quantile: 1st quantile (Q_1) is known as lower quantile and 25% of the data lies below this point.

For 1st quantile, $F(x) = 0.25$

(2) 2nd quantile: 2nd quantile (Q_2) is the median of a data set and 50% of the data lies below this point.

For 2nd quantile, $F(x) = 0.5$

(3) 3rd quantile: 3rd quantile (Q_3) is known as upper quantile and 75% of the data lies below this point.

For 3rd quantile, $F(x) = 0.75$

* Inter quantile range: Interquantile range is the difference between upper quantile and lower quantile.

$$IQR = Q_3 - Q_1$$

Example: Suppose that the diameters of a metal cylinders has a probability density function,

$$f(x) = 1.5 - 6(x - 50.0)^2 \quad \text{for } 49.5 \leq x \leq 50.5$$

- (i) Show that total area under the probability density function = 1 or Prove that this is a valid PDF.
- (ii) Calculate the probability that a metal cylinders has a diameters between 49.8mm and 50.1mm.
- (iii) Find cumulative distribution function.
- (iv) Calculate the expected diameters of a metal cylinders.
- * (v) Find 1st, 2nd & 3rd quartile of a metal cylinders diameters.
- * (vi) Find interquartile range.

Solution:

1st quartile: $F(x) = 0.25$

$$\begin{aligned} (1.5)x - 2(x-50)^3 - 74.5 &= 0.25 \quad [\text{from solution iii}] \\ \Rightarrow 1.5x - 2[x^3 - 150x^2 + 7500x - 125000] - 74.5 - 0.25 &= 0 \\ \Rightarrow 1.5x - 2x^3 + 300x^2 - 15000x + 250000 - 74.75 &= 0 \\ \Rightarrow -2x^3 + 300x^2 - 14998.5x + 249925.25 &= 0 \\ \therefore x &= 50.9 \text{ or } 49.8 \text{ or } 49.2 \end{aligned}$$

Since $49.5 \leq x \leq 50.5$

So 1st quantile = 49.8

Note: Mode set up $\Rightarrow 5$

EQN $\Rightarrow 4$

$-2 \Rightarrow 300 \Rightarrow -14998.5 \Rightarrow 249925.25$

2nd quantile: $F(x) = 0.5$

$(1.5) \times x - 2(x-50)^3 - 74.5 = 0.5$

$\Rightarrow 1.5x - 2[x^3 - 150x^2 + 7500x - 125000] - 74.5 - 0.5 = 0$

$\Rightarrow -2x^3 + 300x^2 - 14998.5x + 250000 - 75 = 0$

$\Rightarrow -2x^3 + 300x^2 - 14998.5x + 249925 = 0$

$\therefore x = 49.1 \text{ or } 50.8 \text{ or } 50$

Since $49.5 \leq x \leq 50.5$

So 2nd quantile = 50

3rd quantile: $F(x) = 0.75$

$(1.5) \times x - 2(x-50)^3 - 74.5 = 0.75$

$\Rightarrow -2x^3 + 300x^2 - 14998.5x + 250000 - 75.25 = 0$

$\Rightarrow -2x^3 + 300x^2 - 14998.5x + 249924.75 = 0$

$\therefore x = 49 \text{ or } 50.7 \text{ or } 50.2$

Since $49.5 \leq x \leq 50.5$

So 3rd quantile = 50.2

Interquartile range: 3rd quantile - 1st quantile

= $50.2 - 49.8$

= 0.4

(Ans.)

Classwork-4: Consider the random variable 'x' with a probability density function of

$$f(x) = \frac{1}{x \ln(1.5)} \quad \text{for } 4 \leq x \leq 6 \quad \text{and} \\ f(x) = 0 \text{ elsewhere.}$$

- (a) Calculate the median.
(b) Find the upper and lower quartiles of this random variable.
(c) What is the interquartile range?

Solution:

(a) Cumulative distribution function,

$$\begin{aligned} & \int_4^x \frac{1}{x \ln(1.5)} dx \\ &= \left[\frac{1}{\ln(1.5)} \ln x \right]_4^x \\ &= \frac{\ln x}{\ln(1.5)} - \frac{\ln 4}{\ln(1.5)} \\ &= \frac{\ln x - \ln 4}{\ln(1.5)} \end{aligned}$$

Median / 2nd quartile: $F(x) = 0.5$

$$\therefore \frac{\ln x - \ln 4}{\ln(1.5)} = 0.5$$

$$\Rightarrow \ln x - \ln 4 = 0.5 \times \ln(1.5)$$

$$\Rightarrow \ln x = [0.5 \times \ln(1.5)] + \ln 4$$

$$\Rightarrow x = e^{[0.5 \times \ln(1.5)] + \ln 4}$$

$$= 4.9$$

(Ans:)

(b) lower / 1st quantile: $F(x) = 0.25$

$$\frac{\ln x - \ln 4}{\ln(1.5)} = 0.25$$

$$\Rightarrow \ln x = [0.25 \times \ln(1.5)] + \ln 4$$

$$\Rightarrow x = e^{[0.25 \times \ln(1.5)] + \ln 4}$$

$$= 4.43$$

Upper / 3rd quantile: $F(x) = 0.75$

$$\frac{\ln x - \ln 4}{\ln(1.5)} = 0.75$$

$$\Rightarrow \ln x = [0.75 \times \ln(1.5)] + \ln 4$$

$$\Rightarrow x = e^{[0.75 \times \ln(1.5)] + \ln 4}$$

$$= 5.42$$

(Ans.)

(c) Interquantile range:

3rd quantile - 1st quantile

$$= 5.42 - 4.43$$

$$= 0.99$$

(Ans.)

Joint Probability Distribution:

The function $f(x, y)$ is a joint probability density function of the continuous random variables X and Y if

$$f(x, y) \geq 0, \text{ for all } (x, y),$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Marginal Probability Distribution:

The marginal distribution of a random variable X is obtained from the joint probability distribution of two random variables X and Y by integrating over the values of the random variable Y . The marginal distribution is the individual probability distribution of the random variable X considered alone. Similarly, the marginal distribution of a random variable Y is obtained from the joint probability distribution of two random variables X and Y by integrating over the values of the random variable X . The marginal distribution is the individual probability distribution of the random variable Y considered alone.

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Conditional Probability:

Let X and Y be two random variables, discrete or continuous, with joint probability distribution $f(x, y)$ and marginal distribution $g(x)$ and $h(y)$, respectively.

The random variables X and Y are said to be statistically independent if and only if

$$f(x, y) = g(x) h(y)$$

Covariance:

The covariance of two random variables X and Y is defined to be

$$\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y)))$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

The covariance can be any positive or negative number, and independent random variables have a covariance of 0.

Correlation:

The correlation between two random variables X and Y is defined to be

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

The correlation takes values between -1 and 1 , and independent random variables have a correlation of 0.

Classwork-5: Suppose that two continuous random variables X and Y have a joint probability density function,

$$f(x, y) = A(x-3)y \quad \begin{matrix} -2 \leq x \leq 3 \\ 4 \leq y \leq 6 \end{matrix}$$

- (a) What is the value of A ? If A is known, show that total area under the probability density function = 1.
- (b) What is $P(0 \leq x \leq 1 \text{ and } 4 \leq y \leq 5)$?
- (c) Construct the marginal probability density function, $g(x)$ and $h(y)$.
- (d) Are the random variables X and Y independent?
- * (e) If $Y=5$, what is the conditional probability density function of X .
- (f) What are the expectation and variance of the random variable X and Y .
- (g) What is the covariance of X and Y .
- (h) What is the correlation between X and Y .

$$\textcircled{a} \int_4^6 \int_{-2}^3 f(x, y) \, dx \, dy = 1$$

$$\Rightarrow \int_4^6 \int_{-2}^3 A(x-3)y \, dx \, dy = 1$$

$$\Rightarrow A \int_4^6 \left[\frac{x^2}{2} - 3x \right]_{-2}^3 y \, dy = 1$$

$$\Rightarrow A \int_4^6 \left[\left(\frac{9}{2} - 9 \right) - \left(\frac{4}{2} + 6 \right) \right] y \, dy = 1$$

$$\Rightarrow A \left(-\frac{25}{2} \right) \left[\frac{y^2}{2} \right]_4^6 = 1$$

$$\Rightarrow A \left(-\frac{25}{2} \right) 10 = 1$$

$$\Rightarrow A(-125) = 1$$

$$\Rightarrow A = -\frac{1}{125} \quad (\text{Ans.})$$

$$\textcircled{b} \, p(0 \leq x \leq 1, 4 \leq y \leq 5)$$

$$\int_4^5 \int_0^1 -\frac{1}{125} (x-3)y \, dx \, dy$$

$$= -\frac{1}{125} \int_4^5 \left[\frac{x^2}{2} - 3x \right]_0^1 y \, dy$$

$$= -\frac{1}{125} \left(-\frac{5}{2} \right) \left[\frac{y^2}{2} \right]_4^5$$

$$= \frac{1}{50} \left[\frac{25}{2} - \frac{16}{2} \right]$$

$$= \frac{1}{50} \times \frac{9}{2}$$

$$= \frac{9}{100} = 0.09 \quad (\text{Ans.})$$

$$\textcircled{c} \, g(x) = \int_4^6 -\frac{1}{125} (x-3)y \, dy$$

$$= -\frac{1}{125} (x-3) \left[\frac{y^2}{2} \right]_4^6$$

$$= -\frac{1}{125} (x-3) \left[\frac{36}{2} - \frac{16}{2} \right]$$

$$= -\frac{1}{125} (x-3) \times 10$$

$$= -\frac{2}{25} (x-3)$$

$$h(y) = \int_{-2}^3 -\frac{1}{125} (x-3)y \, dx$$

$$= -\frac{1}{125} y \left[\frac{x^2}{2} - 3x \right]_{-2}^3$$

$$= -\frac{1}{125} y \left[\left(\frac{9}{2} - 9 \right) - \left(\frac{4}{2} + 6 \right) \right]$$

$$= -\frac{1}{125} y \left(-\frac{25}{2} \right)$$

$$= \frac{y}{10}$$

$$(\text{Ans.})$$

(d) Two random variables are said to be independent if $g(x) \cdot h(y) = f(x, y)$

$$\therefore g(x) \cdot h(y) = -\frac{2}{25}(x-3) \cdot \frac{1}{10}$$

$$= -\frac{1}{125}(x-3)y$$

$$= f(x, y)$$

So, X and Y are independent.

* (e) If $Y=5$, then

$$f(x|y=5) = \frac{f(x, y=5)}{h(y=5)}$$

$$= \frac{-\frac{1}{125}(x-3) \cdot 5}{\frac{5}{10}}$$

$$= -\frac{2}{25}(x-3)$$

$$(f) E(x) = \int_{-2}^3 x g(x) dx$$

$$= \int_{-2}^3 x \left[-\frac{2}{25}(x-3) \right] dx$$

$$= -\frac{2}{25} \int_{-2}^3 [x^2 - 3x] dx$$

$$= -\frac{2}{25} \left[\frac{x^3}{3} - \frac{3x^2}{2} \right]_{-2}^3$$

$$= -\frac{2}{25} \left[\left(-\frac{9}{2}\right) - \left(-\frac{26}{3}\right) \right]$$

$$= -\frac{2}{25} \left[\frac{25}{6} \right]$$

$$= -\frac{1}{3}$$

(Ans.)

$$\begin{aligned}
 E(y) &= \int_4^6 y h(y) dy \\
 &= \int_4^6 y \cdot \frac{y}{10} dy \\
 &= \frac{1}{10} \int_4^6 y^2 dy \\
 &= \frac{1}{10} \left[\frac{y^3}{3} \right]_4^6 \\
 &= \frac{1}{10} \left[\frac{152}{3} \right] \\
 &= \frac{76}{15}
 \end{aligned}$$

$$\begin{aligned}
 E(x^2) &= \int_{-2}^3 x^2 g(x) dx \\
 &= \int_{-2}^3 x^2 \left[-\frac{2}{25}(x-3) \right] dx \\
 &= -\frac{2}{25} \int_{-2}^3 (x^3 - 3x^2) dx \\
 &= -\frac{2}{25} \left[\frac{x^4}{4} - x^3 \right]_{-2}^3 \\
 &= -\frac{2}{25} \left[\left(\frac{27}{4} \right) - (12) \right] \\
 &= \left(-\frac{2}{25} \right) \times \left(-\frac{75}{4} \right) \\
 &= \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 E(y^2) &= \int_4^6 y^2 h(y) dy \\
 &= \int_4^6 y^2 \cdot \frac{y}{10} dy \\
 &= \frac{1}{10} \int_4^6 y^3 dy \\
 &= \frac{1}{10} \left[\frac{y^4}{4} \right]_4^6 \\
 &= \frac{1}{10} [260] \\
 &= 26
 \end{aligned}$$

$$\begin{aligned}
 V(x) &= E(x^2) - (E(x))^2 \\
 &= \frac{3}{2} - \left(-\frac{1}{3} \right)^2 \\
 &= \frac{3}{2} - \frac{1}{9} \\
 &= \frac{25}{18}
 \end{aligned}$$

$$\begin{aligned}
 V(y) &= E(y^2) - (E(y))^2 \\
 &= 26 - \left(\frac{76}{15} \right)^2 \\
 &= 26 - \frac{5 \times 76}{225} \\
 &= \frac{74}{225}
 \end{aligned}$$

(Ans.)

$$(8) \text{Cov}(x, y) = E(xy) - E(x)E(y)$$

$$\text{Here, } E(xy) = \int_4^6 \int_{-2}^3 xy \left[-\frac{1}{125} (x-3)y \right] dx dy$$

$$= -\frac{1}{125} \int_4^6 \int_{-2}^3 (x^2 - 3x) y^2 dx dy$$

$$= -\frac{1}{125} \int_4^6 \left[\frac{x^3}{3} - \frac{3x^2}{2} \right]_{-2}^3 y^2 dy$$

$$= -\frac{1}{125} \int_4^6 \left[\left(-\frac{9}{2} \right) - \left(-\frac{2^6}{3} \right) \right] y^2 dy$$

$$= -\frac{1}{125} \times \frac{25}{6} \int_4^6 y^2 dy$$

$$= -\frac{1}{30} \left[\frac{y^3}{3} \right]_4^6$$

$$= -\frac{1}{30} \left[\frac{152}{3} \right]$$

$$= -\frac{76}{45}$$

$$\therefore \text{Cov}(x, y) = E(xy) - E(x)E(y)$$

$$= -\frac{76}{45} - \left[\left(-\frac{1}{3} \right) \cdot \frac{76}{15} \right]$$

$$= -\frac{76}{45} + \frac{76}{45}$$

$$= 0$$

$$(h) \text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x) \text{Var}(y)}}$$

$$= \frac{0}{\sqrt{\text{Var}(x) \text{Var}(y)}}$$

$$= 0$$

(Ans:)

Joint probability mass function:

Lecture - 9

If the random variables are discrete then the joint probability mass function consists of probability values $P(X=i, Y=j) = P_{ij}$, satisfying

$$1) 0 \leq P_{ij} \leq 1$$

$$2) \sum \sum P_{ij} = 1$$

$X = \text{service time (hrs)}$

	1	2	3	4	
1	0.12	0.08	0.07	0.05	→ 0.32
2	0.08	0.15	0.21	0.13	→ 0.57
3	0.01	0.01	0.02	0.07	→ 0.11
	↓	↓	↓	↓	
	0.21	0.24	0.30	0.25	
					↑ Marginal distribution of Y

↑
Marginal distribution of X

$$P(X=1) = 0.12 + 0.08 + 0.01 = 0.21$$

$$P(Y=1) = 0.12 + 0.08 + 0.07 + 0.05 = 0.32$$

□ Marginal Probability Distributions:

Even though two random variables X and Y may be jointly distributed, if interest is focused on only one of the random variables, then it is appropriate to consider the probability distribution of that random variable alone. This is known as the marginal distribution of the random variable and can be obtained quite simply by summing the joint probability distribution over the values of the other random variable.

For example, for two discrete random variables X and Y , the probability values of the marginal distribution of X are,

$$P(X=i) = \sum_j P_{ij}, \quad i=1,2,3,4 \text{ and } j=1,2,3$$

$$P(X=i) = \sum_{j=1}^3 P_{ij}$$

$$= P_{i1} + P_{i2} + P_{i3}$$

$$P(X=1) = \sum_{j=1}^3 P_{1j}$$

$$= P_{11} + P_{12} + P_{13}$$

Conditional Probability Distributions:

If two random variables X and Y are jointly distributed, then it is sometimes useful to consider the distribution of one random variable conditional on the other random variable having taken a particular value.

$$P(X|Y=j) = \frac{P(X, Y=j)}{P(Y=j)} \quad [\text{chapter-8, Q(2)}]$$

$$P(Y|X=j) = \frac{P(Y, X=j)}{P(X=j)}$$

Covariance:

$$\text{Cov}(X, Y) = E(X, Y) - E(X)E(Y)$$

Correlation:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}}$$

$$[-1 \text{ to } 1]$$

Question: A company that services air conditioners units in residences. If the random variable X , taking the values 1, 2, 3 & 4, is the service time in hours taken at a particular location, and the random variable Y , taking the values 1, 2, and 3, is the numbers of air conditioner units at the location, then these two random variables can be thought of as jointly related.

Suppose that their joint probability mass function P_{ij} is given in the following figure. This figure indicates, for example, that there is a probability of 0.12 that $X=1$ and $Y=1$, so that there is a probability of 0.12 that a particular location chosen at random has one air conditioner unit that takes a technician one hour to service. Similarly, there is a probability of 0.07 that a location has three air conditioner units that take four hours to service.

$X = \text{service time (hrs)}$

$Y = \text{numbers of air conditioner units}$

	1	2	3	4
1	0.12	0.08	0.07	0.05
2	0.08	0.15	0.21	0.13
3	0.01	0.01	0.02	0.07

Y = number of
air conditioners units

X = service time (hrs)

j	1	2	3	4 $\rightarrow i$
1	0.12	0.08	0.07	0.05
2	0.08	0.15	0.21	0.13
3	0.01	0.01	0.02	0.07

- ① Show that this is a valid joint probability mass function.

Solution:

$$\sum_{i=1}^4 \sum_{j=1}^3 P_{ij} = (0.12 + 0.08 + 0.01 + 0.08 + 0.15 + 0.01 + 0.07 + 0.21 + 0.02 + 0.05 + 0.13 + 0.07) = 1$$

- ② Calculate the probability that a location has no more than two air conditioners units that take no more than two hours to service.

$$i = 1, 2; j = 1, 2$$

Solution:

$$P_{11} + P_{12} + P_{21} + P_{22} = 0.12 + 0.08 + 0.08 + 0.15 = 0.43$$

③ Find marginal probability mass function.

Solution: Marginal distribution of X

$$P(X=1) = 0.12 + 0.08 + 0.01 = 0.21$$

$$P(X=2) = 0.08 + 0.15 + 0.01 = 0.24$$

$$P(X=3) = 0.07 + 0.21 + 0.02 = 0.3$$

$$P(X=4) = 0.05 + 0.13 + 0.07 = 0.25$$

Marginal distribution of Y

$$P(Y=1) = 0.12 + 0.08 + 0.07 + 0.05 = 0.32$$

$$P(Y=2) = 0.08 + 0.15 + 0.21 + 0.13 = 0.57$$

$$P(Y=3) = 0.01 + 0.01 + 0.02 + 0.07 = 0.11$$

(Ans.)

④ Calculate the expected service time and the variance in the service times.

Solution: We know,

$$E(X) = \sum x p(x) \quad [\text{For discrete random variables}]$$

$$E(X^2) = \sum x^2 p(x)$$

∴ The expected service time,

$$E(X) = \sum_{i=1}^4 i P(X=i)$$

$$= 1 P(X=1) + 2 P(X=2) + 3 P(X=3) + 4 P(X=4)$$

$$= (1 \times 0.21) + (2 \times 0.24) + (3 \times 0.3) + (4 \times 0.25)$$

$$= 2.59$$

Now,

$$E(X^2) = \sum_{i=1}^4 i^2 P(X=i)$$

$$= 1^2 P(X=1) + 2^2 P(X=2) + 3^2 P(X=3) + 4^2 P(X=4)$$

$$= (1 \times 0.21) + (4 \times 0.24) + (9 \times 0.3) + (16 \times 0.25)$$

$$= 7.87$$

∴ The variance in service times,

$$V(X) = E(X^2) - (E(X))^2$$

$$= 7.87 - (2.59)^2$$

$$= 1.162$$

(Ans:)

⑤ Calculate the expected number of units serviced and the variance in the number of units serviced.

Solution: We know,

$$E(Y) = \sum Y P(Y)$$

$$E(Y^2) = \sum Y^2 P(Y)$$

∴ The expected number of units

$$E(Y) = \sum_{j=1}^3 j P(Y=j)$$

$$= 1 P(Y=1) + 2 P(Y=2) + 3 P(Y=3)$$

$$= (1 \times 0.32) + (2 \times 0.57) + (3 \times 0.11)$$

$$= 1.79$$

Now,

$$E(Y^2) = \sum_{j=1}^3 j^2 P(Y=j)$$

$$= 1^2 P(Y=1) + 2^2 P(Y=2) + 3^2 P(Y=3)$$

$$= (1 \times 0.32) + (4 \times 0.58) + (9 \times 0.11)$$

$$= 3.59$$

∴ The variance in the numbers of units serviced

$$V(Y) = 3.59 - (1.79)^2$$

$$= 0.3859$$

(Ans:)

⑥ Suppose that a technician is visiting a location that is known to have three air conditioners units, what is the conditional distribution of the service time X .

Solution: We know,

$$P(X|Y=j) = \frac{P(X, Y=j)}{P(Y=j)}$$

$$\therefore P(X|Y=3) = \frac{P(X, Y=3)}{P(Y=3)}$$

$$\text{Here, } P(Y=3) = 0.01 + 0.01 + 0.02 + 0.07 = 0.11$$

$$\therefore P(X=1|Y=3) = \frac{P(X=1, Y=3)}{P(Y=3)} = \frac{0.01}{0.11} = 0.091$$

$$P(X=2|Y=3) = \frac{P(X=2, Y=3)}{P(Y=3)} = \frac{0.01}{0.11} = 0.091$$

$$P(X=3|Y=3) = \frac{P(X=3, Y=3)}{P(Y=3)} = \frac{0.02}{0.11} = 0.182$$

$$P(X=4|Y=3) = \frac{P(X=4, Y=3)}{P(Y=3)} = \frac{0.07}{0.11} = 0.636 \quad (\text{Ans.})$$

⑦ Calculate the conditional expectation of the service time.

Solution:

$$\begin{aligned} E(X|Y=3) &= \sum_{i=1}^4 i P(X=i|Y=3) \\ &= 1 P(X=1|Y=3) + 2 P(X=2|Y=3) + 3 P(X=3|Y=3) \\ &\quad + 4 P(X=4|Y=3) \\ &= (1 \times 0.091) + (2 \times 0.091) + (3 \times 0.182) + (4 \times 0.636) \\ &= 3.363 \end{aligned}$$

(Ans.)

* Extra: Conditional expectation of the number of units.

$$\begin{aligned} E(Y|X=3) &= \sum_{j=1}^3 j P(Y=j|X=3) \\ &= (1 \times 0.23) + (2 \times 0.7) \\ &\quad + (3 \times 0.067) \\ &= 1.831 \end{aligned}$$

Here,

$$P(X=3) = 0.3$$

$$\therefore P(Y=1|X=3) = \frac{0.07}{0.3} = 0.23$$

$$P(Y=2|X=3) = \frac{0.21}{0.3} = 0.7$$

$$P(Y=3|X=3) = \frac{0.02}{0.3} = 0.067$$

⑧ Calculate the covariance and correlation.

Solution:

The expected service time is $E(X) = 2.59$

The expected number of units serviced is $E(Y) = 1.79$

$$\text{Now, } E(X, Y) = \sum_{i=1}^4 \sum_{j=1}^3 ij P(X=i, Y=j)$$

$$\begin{aligned} &= (1 \times 1 \times 0.12) + (1 \times 2 \times 0.08) + (1 \times 3 \times 0.01) \\ &\quad + (2 \times 1 \times 0.08) + (2 \times 2 \times 0.15) + (2 \times 3 \times 0.01) \\ &\quad + (3 \times 1 \times 0.07) + (3 \times 2 \times 0.21) + (3 \times 3 \times 0.02) \\ &\quad + (4 \times 1 \times 0.05) + (4 \times 2 \times 0.13) + (4 \times 3 \times 0.07) \\ &= 0.31 + 0.82 + 1.65 + 2.08 \\ &= 4.86 \end{aligned}$$

$$\begin{aligned} \therefore \text{Cov}(X, Y) &= E(X, Y) - E(X)E(Y) \\ &= 4.86 - (2.59)(1.79) \\ &= 0.2239 \end{aligned}$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}} = \frac{0.2239}{\sqrt{1.162 \times 0.3859}} = 0.334$$

(Ans)

Bernoulli distribution:

Lecture-10 & 11

A single trial is conducted which takes a binary outcome.

The probability mass function is

$$P(X=x) = p^x (1-p)^{1-x} \quad x=0,1$$

$$\text{Expectation: } E(x) = p$$

$$\text{Variance: } V(x) = p(1-p)$$

Example: If you toss a coin once, then the random variable X indicates two possible outcomes that is 1 & 0, of the head (H).


$$\left. \begin{array}{l} P(X=1) = \theta \\ P(X=0) = 1-\theta \end{array} \right\}$$
$$P(X=k) = \theta^k (1-\theta)^{1-k}$$

if $x=0$, $P(X=0) = \theta^0 (1-\theta)^{1-0} = 1-\theta$

if $x=1$, $P(X=1) = \theta^1 (1-\theta)^{1-1} = \theta$

Here, when we get head (H) from flipping a coin, then $X=1$. If we don't get H, then $X=0$.

Binomial distribution:

n trials ($n > 1$) are conducted which takes a binary/two outcome. The probability mass function is

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

$$\text{Expectation: } E(x) = np$$

$$\left[\binom{n}{x} = {}^nC_x \right]$$

$$\text{Variance: } V(x) = np(1-p)$$

Example: If you toss a coin 5 times and the random variable X indicates the number of tails, then

$$n=5, \quad x=0, 1, 2, 3, 4, 5.$$

Here, the maximum number of tails we can get is 5 and the minimum is 0.

Example: Suppose a milk factory has 20 containers and there is a probability of 0.261 that a milk container is underweight.

- What is the distribution of the numbers of underweight containers in a box?
- Calculate expected numbers of underweight cartons in a box and also calculate its variance.
- Calculate the probability that a box contains exactly seven underweight containers.

(d) Calculate the probability that a box contains no more than three underweight containers.

(e) Calculate the probability that a box contains at least two underweight containers.

Solution:

(a) Binomial distribution

We know, $\boxed{P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}}$ $x = 0, 1, 2, \dots, 20$

$$\therefore P(X=x) = \binom{20}{x} p^x (1-p)^{20-x}$$

(b) Expected number of underweight containers in a box is,

$$E(x) = np$$

$$= 20 \times 0.261$$

$$= 5.22$$

$$\left. \begin{array}{l} n=20 \\ p=0.261 \end{array} \right|$$

$$\text{Variance, } V(x) = np(1-p)$$

$$= 20 \times 0.261 \times (1 - 0.261)$$

$$= 3.86$$

(c) If the box contains exactly 7 underweight containers, then,

$$P(X=7) = \binom{20}{7} 0.261^7 (1 - 0.261)^{20-7}$$

$$= 0.125$$

(d) If the box contains no more than three underweight containers, then,

$$\begin{aligned}P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\&= \binom{20}{0} 0.261^0 (1-0.261)^{20-0} + \binom{20}{1} 0.261^1 (1-0.261)^{20-1} \\&\quad + \binom{20}{2} 0.261^2 (1-0.261)^{20-2} + \binom{20}{3} 0.261^3 (1-0.261)^{20-3} \\&= (2.36 \times 10^{-3}) + 0.0167 + 0.0559 + 0.1185 \\&= 0.19346\end{aligned}$$

(e) If the box contains at least two underweight containers, then,

$$P(X \geq 2) = P(X=2) + P(X=3) + \dots + P(X=20)$$

We know that, total probability = 1

$$\therefore P(X=0) + P(X=1) + P(X=2) + P(X=3) + \dots + P(X=20) = 1$$

$$\Rightarrow P(X=2) + P(X=3) + \dots + P(X=20) = 1 - P(X=0) - P(X=1)$$

$$= 1 - (2.36 \times 10^{-3}) - 0.0167$$

$$= 0.98094$$

Poisson distribution:

The Poisson distribution is used when a random variable counts the (numbers of) events that occur in an time interval. For example,

(1) The numbers of telephone calls received by an operators within a certain time limit.

(2) The number of patients arriving in an emergency room between 10 pm and 11 pm.

The probability mass function is

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, 3, \dots$$

$$\text{Expectation: } E(X) = \lambda$$

$$\text{Variance: } V(X) = \lambda$$

Difference between binomial distribution and Poisson distribution:

(1) In binomial distribution, numbers of trials are fixed. In poisson distribution, numbers of trials are infinite.

(2) In Binomial distribution, $\text{Mean} \geq \text{Variance}$. In Poisson distribution,
 $\text{Mean} = \text{Variance}$.

For Binomial distribution, Expectation: $E(x) = np$

Variance: $V(x) = np(1-p)$ [$0 \leq p \leq 1, n > 1$]

So, $\text{Mean} = \text{Variance}$

When $p=0$

$$\Rightarrow np = np(1-0)$$

$$\Rightarrow n \cdot 0 = n \cdot 0(1-0)$$

$$\Rightarrow 0 = 0$$

Poisson distribution

$\text{Mean}(n) > \text{Variance}(0)$

When $p=1$

Now, when $p=1$,

$$\begin{aligned} \text{Mean} &= n \cdot 1 \\ &= n(n > 1) \end{aligned}$$

$$\begin{aligned} \text{Variance} &= n \cdot 1(1-1) \\ &= 0 \end{aligned}$$

So, $\text{Mean} > \text{Variance}$ [$0 < p < 1$]

$$\Rightarrow np > np(1-p)$$

Example: $50 > 50(1-p)$

Hence,

$$0 < (1-p) < 1$$

Let, $p = 0.2$

$$\therefore 50 > 50(1-0.2)$$

$$\Rightarrow 50 > 50 \times 0.8$$

$$\Rightarrow 50 > 40$$

(3) Example of binomial distribution,

→ Tossing a coin experiment.

Example of Poisson distribution,

→ The number of patients arriving in an emergency room between 10 pm to 11 pm.

Example: Suppose that the number of errors in a piece of software has a parameter $\lambda = 3$. This parameter immediately implies that the expected number of errors is three and that the variance in the number of errors is also equal to three.

(a) What is the distribution of the number of errors in a piece of software.

(b) Calculate the probability that a piece of software has no errors.

(c) Calculate the probability that there are three or more errors in a piece of software.

Solution:

(a) Here, Expectation = Variance

So, the number of errors in the piece of software follows Poisson distribution.

We know, $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$ $x = 0, 1, 2, 3, \dots$

$$\therefore P(X=x) = \frac{e^{-3} 3^x}{x!}$$

(b) If the software has no errors, then

$$P(X=0) = \frac{e^{-3} 3^0}{0!}$$

$$= 0.05$$

(c) If the software has three or more errors, then

$$P(X \geq 3) = P(X=3) + P(X=4) + \dots = ?$$

We know,

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + \dots = 1$$

$$\Rightarrow P(X=3) + P(X=4) + \dots = 1 - P(X=0) - P(X=1) - P(X=2)$$

$$= 1 - \frac{e^{-3} 3^0}{0!} - \frac{e^{-3} 3^1}{1!} - \frac{e^{-3} 3^2}{2!}$$

$$= 1 - 0.05 - 0.15 - 0.224$$

$$= 0.576 \quad (\text{Ans.})$$

Q Geometric Distribution:

The binomial distribution is the distribution of the number of successes occurring in a fixed number of trials n , it is sometimes of interest to count the number of trials performed until the first success occurs. Such a random variable is said to have a geometric distribution.

The probability mass function is

$$P(X=x) = (1-p)^{(x-1)} p \quad x=1, 2, 3, \dots$$

$$\text{Expectation } E(X) = \frac{1}{p}$$

$$\text{Variance } V(X) = \frac{1-p}{p^2}$$

Example: Suppose that a company wishes to hire one new workers and that each applicant interviewed has a probability of 0.6 of being found acceptable.

- (1) What is the distribution of the total numbers of applicants that the company needs to interview?
- (2) Calculate the probability that exactly six applicants needed to be interviewed.
- (3) Calculate the probability that the company allows up to/at most six applicants to be interviewed.
- (4) Calculate the probability that at least six applicants needed to be interviewed.
- (5) Calculate the expected numbers of interviews.

Solution:

(1) Here, the company wishes to hire one new worker. So, the total numbers of applicants that the company needs to interview follows geometric distribution.

We know, $P(X=x) = (1-p)^{(x-1)} p$

$x = 1, 2, 3, \dots$

$\therefore$ The p.m.f. is, $P(X=x) = (1-0.6)^{(x-1)} (0.6)$

→ [numbers of applicants needed to be interviewed]

(2) Probability that exactly six applicants need to be interviewed is,

$$\begin{aligned}P(X=6) &= (1-0.6)^{(6-1)}(0.6) \\&= 6.144 \times 10^{-3} \\&= 0.006144\end{aligned}$$

(3) Probability that up to six applicants will be allowed to be interviewed is,

$$\begin{aligned}P(X \leq 6) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\&\quad + P(X=5) + P(X=6) \\&= (1-0.6)^{(1-1)}(0.6) + (1-0.6)^{(2-1)}(0.6) + (1-0.6)^{(3-1)}(0.6) \\&\quad + (1-0.6)^{(4-1)}(0.6) + (1-0.6)^{(5-1)}(0.6) + (1-0.6)^{(6-1)}(0.6) \\&= 0.6 + 0.24 + 0.096 + 0.0384 \\&\quad + 0.01536 + 0.006144 \\&= 0.995904\end{aligned}$$

(4) Probability that at least six applicants need to be interviewed is

$$P(X \geq 6) = P(X=6) + P(X=7) + \dots = ?$$

We know,

$$P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7) + \dots = 1$$

$$\begin{aligned} \Rightarrow P(X=6) + P(X=7) + \dots &= 1 - P(X=1) - P(X=2) - P(X=3) - P(X=4) - P(X=5) \\ &= 1 - 0.6 - 0.24 - 0.096 - 0.0384 - 0.01536 \\ &= 0.01024 \end{aligned}$$

(5) The expected number of interviews,

$$\begin{aligned} E(X) &= \frac{1}{p} \\ &= \frac{1}{0.6} \\ &= 1.67 \end{aligned}$$

② Negative Binomial Distribution:

The binomial distribution is the distribution of the number of successes occurring in a fixed number of trials n , it is sometimes of interest to count the number of trials performed until the r th ($r > 1$) success occurs. Such a random variable is said to have a negative binomial distribution.

The probability mass function (p.m.f.) is

$$P(X=x) = \binom{x-1}{r-1} (1-p)^{(x-r)} p^r$$

$$x=r, r+1, r+2, \dots$$

$$\text{Expectation } E(X) = \frac{r}{p}$$

$$\text{Variance } V(X) = \frac{r(1-p)}{p^2}$$

Example: Suppose that a company wishes to hire

$n=3$ three new workers and that each applicant interviewed has a probability of 0.6 of being found acceptable.

- (1) What is the distribution of the total numbers of applicants that the company needs to interview?
- (2) Calculate the probability that exactly six applicants need to be interviewed.
- (3) Calculate the probability that the company allows up to / at most six applicants to be interviewed.
- (4) Calculate the probability that at least six applicants need to be interviewed.
- (5) Calculate the expected number of interviews.

Solution:

(1) We know, $P(X=x) = \binom{x-1}{n-1} (1-p)^{x-n} p^n$

$\therefore$ The p.m.f. is, $P(X=x) = \binom{x-1}{3-1} (1-0.6)^{x-3} p^3$ $x = 3, 4, 5, \dots$

[at least three new workers]

(2) Probability that exactly six applicants need to be interviewed is,

$$\begin{aligned}P(X=6) &= \binom{6-1}{3-1} (1-0.6)^{(6-3)} (0.6)^3 \\&= \binom{5}{2} (1-0.6)^{(6-3)} (0.6)^3 \\&= 0.138\end{aligned}$$

(3) Probability that upto six applicants will be allowed to be interviewed is,

$$\begin{aligned}P(X \leq 6) &= P(X=3) + P(X=4) + P(X=5) + P(X=6) \\&= \binom{3-1}{3-1} (1-0.6)^{(3-3)} (0.6)^3 + \binom{4-1}{3-1} (1-0.6)^{(4-3)} (0.6)^3 \\&\quad + \binom{5-1}{3-1} (1-0.6)^{(5-3)} (0.6)^3 + \binom{6-1}{3-1} (1-0.6)^{(6-3)} (0.6)^3 \\&= 0.216 + 0.2592 + 0.20736 + 0.13824 \\&= 0.8208\end{aligned}$$

(4) Probability that at least six applicants need to be interviewed is,

$$P(X \geq 6) = P(X=6) + P(X=7) + \dots = ?$$

We know,

$$P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7) + \dots = 1$$

$$\Rightarrow P(X=6) + P(X=7) + \dots = 1 - P(X=3) - P(X=4) - P(X=5)$$

$$= 1 - 0.216 - 0.2592 - 0.20736$$

$$= 0.31744$$

(5) Expected number of interviews,

$$E(X) = \frac{n}{p}$$

$$= \frac{3}{0.6}$$

$$= 5$$

(Ans:)

Q Normal Distribution: (Continuous random variable)

The probability density function of normal distribution is,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$-\infty < x < \infty$$

$$-\infty < \mu < \infty$$

$$0 \leq \sigma^2 < \infty$$

$$\text{Expectation: } E(x) = \mu$$

$$\text{Variance: } V(x) = \sigma^2$$

Q Standard normal distribution:

When $\mu=0$ and $\sigma^2=1$ then the normal distribution is called standard normal distribution.

The probability density function of standard normal distribution is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < \infty$$

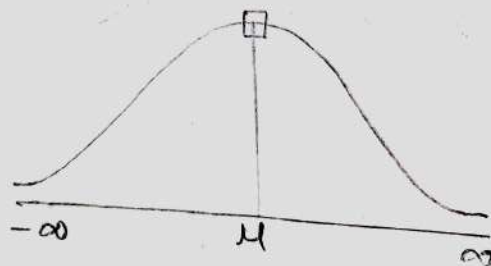
Q Property of a normal distribution:

[1] It is symmetric.

[2] Mean = Mode = Median

Same {

- Mean = The expected/average value of x .
- Mode = The peak point.
- Median = The middle point.



③ It is unimodal.


unimodal


bimodal


multimodal

④ The total area under the curve is equal to one.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

⑤ The normal curve approaches, but never touches, the x-axis.

Extra:

Transformation:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}z^2} \sigma dz$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

Let,

$$z = \frac{x-\mu}{\sigma}$$

$$\Rightarrow z = \frac{x}{\sigma} - \frac{\mu}{\sigma}$$

$$\Rightarrow \frac{dz}{dx} = \frac{1}{\sigma}$$

$$\Rightarrow dz = \frac{1}{\sigma} dx$$

$$\Rightarrow \sigma dz = dx$$

That is if $X \sim N(\mu, \sigma^2)$ and if you want to transform the normal distribution to standard normal distribution then the transform random variable is $z = \frac{x-\mu}{\sigma}$ (z score)

x	$-\infty$	∞
z	$-\infty$	∞

⊞ Probability Calculation for Normal Distribution;

⊞

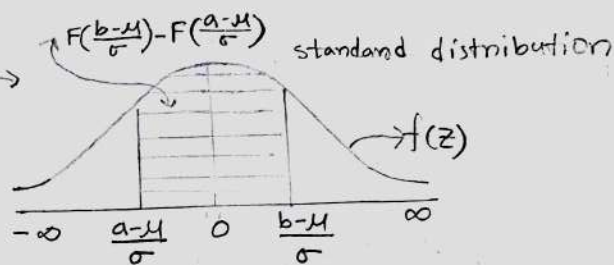
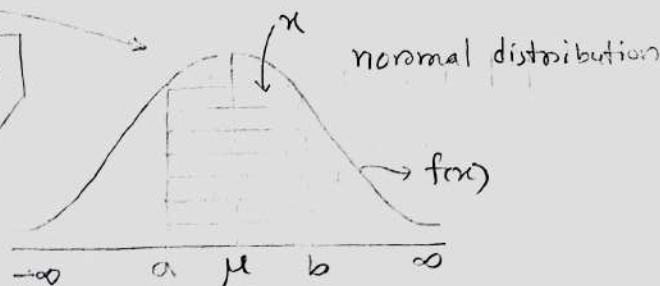
$$P(a < x < b)$$

$$= P\left(\frac{a-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right)$$

$$= P\left(\frac{a-\mu}{\sigma} < z < \frac{b-\mu}{\sigma}\right)$$

$$= F\left(\frac{b-\mu}{\sigma}\right) - F\left(\frac{a-\mu}{\sigma}\right)$$

extracted value of x
 $\rightarrow F(x)$
 plug in $z = \frac{x-\mu}{\sigma}$
 $\downarrow$
 $\sqrt{V(x)}$
 square root of variance of x



so, if we subtract $-\infty$ to
 white part from $-\infty$ to
 the shaded part, we get
 the area of the shaded
 part.

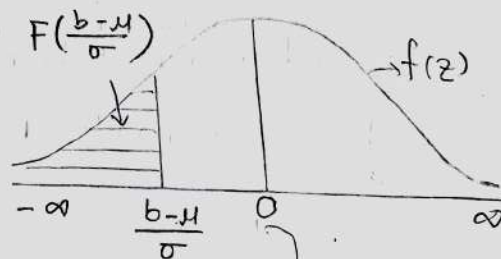
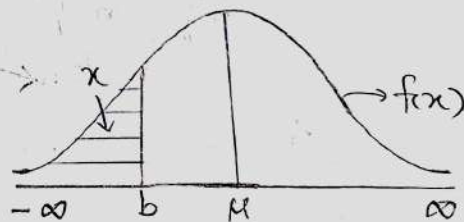
⊞

$$P(x < b)$$

$$= P\left(\frac{x-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right)$$

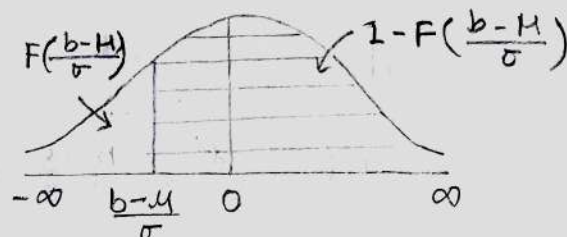
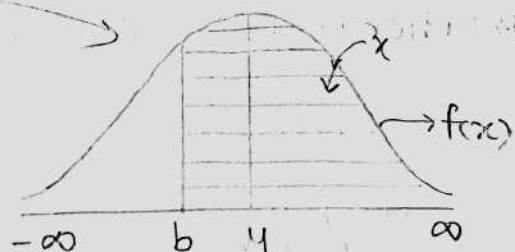
$$= P\left(z < \frac{b-\mu}{\sigma}\right)$$

$$= F\left(\frac{b-\mu}{\sigma}\right)$$



In standard normal distribution
 mean = 0, variance = 1.
 μ σ^2

$$\begin{aligned}
 & P(x > b) \\
 &= P\left(\frac{x - \mu}{\sigma} > \frac{b - \mu}{\sigma}\right) \\
 &= P(z > \frac{b - \mu}{\sigma}) \\
 &= 1 - F\left(\frac{b - \mu}{\sigma}\right)
 \end{aligned}$$



Example:

A company manufactures concrete blocks that are used for construction purpose. Suppose that the weight of the individual concrete blocks are normally distributed with a mean value of $\mu = 11 \text{ kg}$ and a standard deviation of $\sigma = 0.3 \text{ kg}$.

- i) Calculate the probability that a concrete block weight is less than 10.5 kg.
- ii) Calculate the probability that a concrete block weight is within 10 kg to 12 kg.
- iii) Calculate the probability that a concrete block weight is greater than 10.5 kg.

Solution:

(i) Concrete block weight is less than 10.5 kg.

$$\mu = 11 \text{ kg}$$

$$\sigma = 0.3 \text{ kg.}$$

$$\therefore P(X < 10.5)$$

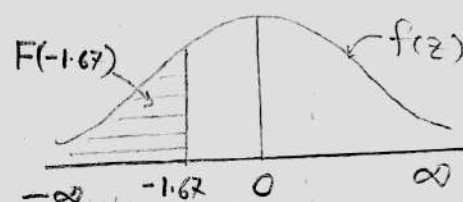
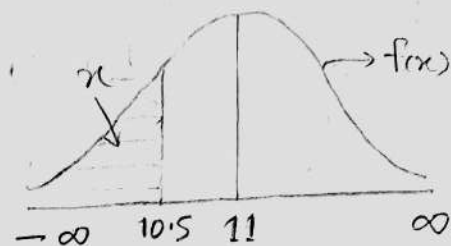
$$= P\left(\frac{X - \mu}{\sigma} < \frac{10.5 - \mu}{\sigma}\right)$$

$$= P(Z < \frac{10.5 - 11}{0.3})$$

$$= P(Z < -1.67)$$

$$= F(-1.67)$$

$$= 0.0475$$



(ii) Concrete block weight is within 10 kg to 12 kg.

$$P(10 < X < 12)$$

$$= P\left(\frac{10 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{12 - \mu}{\sigma}\right)$$

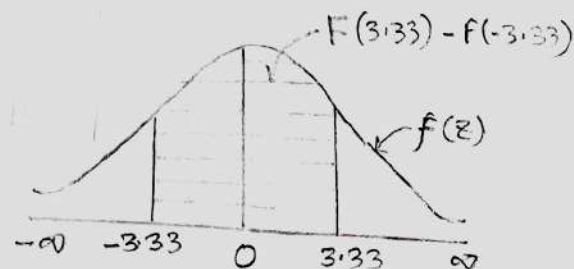
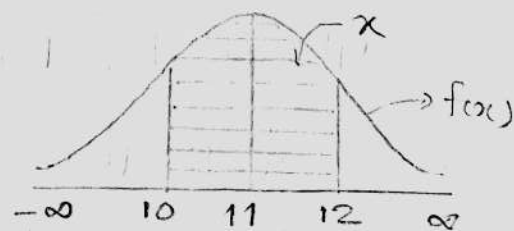
$$= P\left(\frac{10 - 11}{0.3} < Z < \frac{12 - 11}{0.3}\right)$$

$$= P(-3.33 < Z < 3.33)$$

$$= F(3.33) - F(-3.33)$$

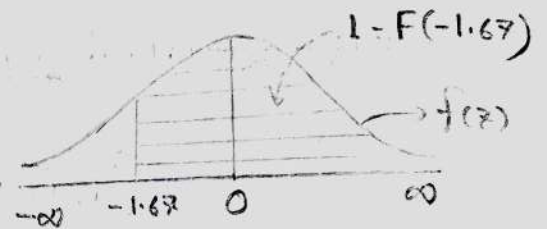
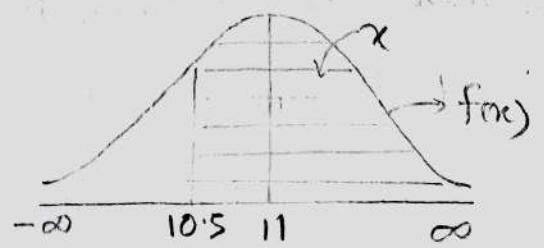
$$= 0.9996 - 0.0004$$

$$= 0.9992$$



(iii) Concrete block weight is greater than 10.5 kg.

$$\begin{aligned} P(X > 10.5) &= P\left(\frac{X-M}{\sigma} > \frac{10.5-M}{\sigma}\right) \\ &= P\left(Z > \frac{10.5-11}{0.3}\right) \\ &= P(Z > -1.67) \\ &= 1 - F(-1.67) \\ &= 1 - 0.0475 \\ &= 0.9525 \end{aligned}$$



(Ans:)

* if less than $-3.49 \rightarrow 0$
 $+3.49 \rightarrow 1$

Central limit theorem:

Lecture - 13

When sample size is large (≥ 30), the average of a set of independent identically distributed random variables ($\bar{X}$) is always approximately normally distributed with mean = population mean and variance = population variance divided by sample size.

$$X \sim \text{Poi}(\lambda), E(X) = \lambda, V(X) = \lambda$$

[X follows Poisson distribution]

↓
Popⁿ mean

↓
Popⁿ variance

$$n \geq 30, \bar{X} \sim N \left(\text{Pop}^n \text{ mean}, \frac{\text{Pop}^n \text{ variance}}{n} \right)$$

[X follows normal distribution]

$$\bar{X} \sim N \left(\lambda, \frac{\lambda}{n} \right)$$

$\bar{X}$ = average value of the random variable

N = Normal distribution

n = sample space

μ = population mean = $E(\bar{X})$

σ^2 = population variance = $V(\bar{X})$

$\sim$ = Follows

When n is large,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ --- Avg.}$$

and it indicates that,

$$E(\bar{X}) = \mu \text{ --- Avg.} \rightarrow \text{Keyword: average} \quad \text{Population mean}$$

$$\Rightarrow E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \mu$$

$$\Rightarrow \frac{1}{n} E(X_1 + X_2 + \dots + X_n) = \mu \quad [\text{Formula, } E(cX) = cE(X)]$$

$$\Rightarrow E(X_1 + X_2 + \dots + X_n) = n\mu$$

$$\Rightarrow E\left(\sum_{i=1}^n X_i\right) = n\mu \text{ --- Total --- Keyword: Total}$$

$$V(\bar{X}) = \frac{\sigma^2}{n} \text{ --- Avg.} \rightarrow \text{Keyword: average} \quad \text{Population variance}$$

$$\Rightarrow V\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{\sigma^2}{n}$$

$$\Rightarrow \frac{1}{n^2} V(X_1 + X_2 + \dots + X_n) = \frac{\sigma^2}{n} \quad [\text{Formula } V(cX) = c^2 V(X)]$$

$$\Rightarrow V(X_1 + X_2 + \dots + X_n) = n^2 \cdot \frac{\sigma^2}{n}$$

$$\Rightarrow V(X_1 + X_2 + \dots + X_n) = n\sigma^2$$

$$\Rightarrow V\left(\sum_{i=1}^n X_i\right) = n\sigma^2 \text{ --- Total --- Keyword: Total}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$\begin{aligned} V(cX) &= E((cX)^2) - (E(cX))^2 \\ &= c^2 E(X^2) - c^2 (E(X))^2 \\ &= c^2 (E(X^2) - (E(X))^2) \\ &= c^2 V(X) \end{aligned}$$

Summation of X , [Keyword: Total]

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2) \text{ --- Total}$$

Example: The number of flaws in a glass sheet has a Poisson distribution with a parameter $\lambda = 0.5$.

- (a) What is the distribution of the average number of flaws per sheet in 100 sheets of glass?
- (b) Calculate the probability that this average is between 0.45 and 0.55.
- (c) What is the distribution of the total number of flaws in 100 sheets of glass?
- (d) Calculate the probability that there are fewer than 40 flaws in 100 sheets of glass? (total)

Solution: Here, the given problem follows Poisson distribution.

Now, The probability mass function of Poisson distribution is,

$$P(X=x) = \frac{e^{-0.5} 0.5^x}{x!}$$

$$x = 0, 1, 2, 3, \dots$$

↳ i.e. 0 flaw in a glass

$$\text{Expectation } E(X) = 0.5$$

$$\text{Variance } V(X) = 0.5$$

(a) The distribution of the average numbers of flaws per sheet in 100 sheets of glass follows normal distribution with mean, $\mu = 0.5$, and variance, $\sigma^2 = \frac{0.5}{100}$.

$$\therefore \boxed{\bar{X} \sim N(0.5, \frac{0.5}{100})} \checkmark$$

[Here, the sample size is
larger than 30, $100 > 30$]

(b) average is between 0.45 and 0.55.

$$P(0.45 < \bar{X} < 0.55)$$

$$= P\left(\frac{0.45 - \mu}{\sigma} < \frac{\bar{X} - \mu}{\sigma} < \frac{0.55 - \mu}{\sigma}\right)$$

$$= P\left(\frac{0.45 - 0.5}{\sqrt{\frac{0.5}{100}}} < Z < \frac{0.55 - 0.5}{\sqrt{\frac{0.5}{100}}}\right)$$

$$= P(-0.707 < Z < 0.707)$$

$$= F(0.707) - F(-0.707)$$

$$* = F(0.71) - F(-0.71)$$

$$= 0.7611 - 0.2389$$

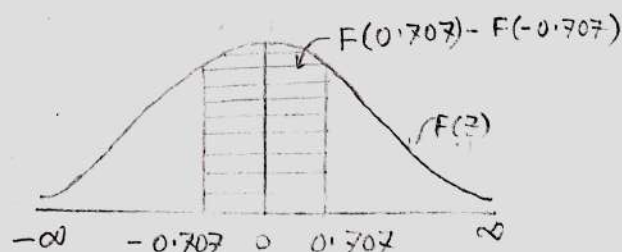
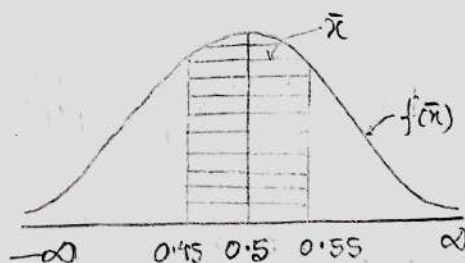
$$= 0.5222$$

Here,

$$\mu = 0.5$$

$$\sigma^2 = \frac{0.5}{100}$$

$$\sigma = \sqrt{\frac{0.5}{100}}$$



(c) The distribution of the total numbers of flaws in 100 sheets of glass

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$$

$$\sum_{i=1}^{100} X_i \sim N(100 \times 0.5, 100 \times 0.5)$$

Hence,

$$n = 100$$

$$\mu = 0.5$$

$$\sigma^2 = 0.5$$

$$\Rightarrow \sum_{i=1}^{100} X_i \sim N(50, 50) \quad \checkmark$$

$$X_1 + X_2 + \dots + X_{100} \sim N(50, 50)$$

(d) Probability that there are fewer than 40 flaws in 100 sheets of glass is calculated below,

$$P\left(\sum_{i=1}^{100} X_i < 40\right)$$

$$= P\left(\frac{\sum_{i=1}^{100} X_i - \mu'}{\sigma'} < \frac{40 - \mu'}{\sigma'}\right)$$

$$= P\left(\frac{\sum_{i=1}^{100} X_i - 50}{\sqrt{50}} < \frac{40 - 50}{\sqrt{50}}\right)$$

$$= P(Z < -1.41)$$

$$= F(-1.41)$$

$$= 0.0793$$

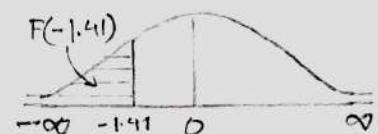
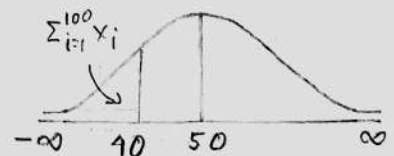
$$\mu' = 100 \times 0.5$$

$$= 50$$

$$\sigma'^2 = 100 \times 0.5$$

$$= 50$$

$$\sigma' = \sqrt{50}$$

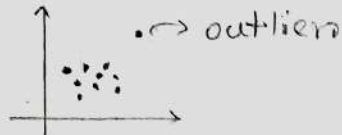


(Ans.)

Table I: Cumulative Distribution Function of the Standard Normal Distribution

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0061
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5339
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9019	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9610	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998



❑ Population: A population consists of all observation in an experiment.

❑ Sample: A sample is a subset of the population.

❑ Random sample: A random sample is one in which the elements of the sample are chosen at random from the population.

❑ Difference between bar charts and histograms:

① Bar charts are used to represent categorical data. Histograms are used to represent numerical or continuous data.

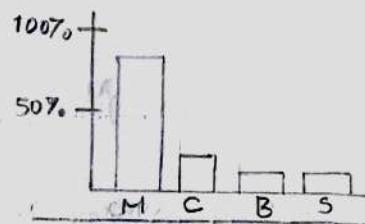
② In bar charts, bars do not touch each other, hence there are spaces between bars.

In histograms, bars touch each other, hence there are no spaces between bars.

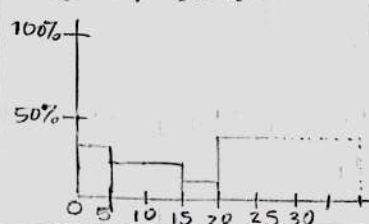
③ In bar charts, width of bars need to be same.

In histograms, width of bars don't need to be same.

Example: M: 80% C: 15%
B: 2.5% S: 2.5%



(0-5): 30% (5-15)= 25%, (15-20)= 10%
(20>): 35%



** ④ An outlier is an observation point that is distant from others observations in a data set.

→ This method is used to estimate the parameter of an unknown quantity in a distribution.

Q1 Maximum likelihood method:

Step 1: Likelihood function $L(\theta) = \prod_{i=1}^m f(x_i; \theta)$

Step 2: $\ln L(\theta)$

Step 3: $\frac{\partial}{\partial \theta} \ln L(\theta)$

Step 4: $\frac{\partial}{\partial \theta} \ln L(\theta) = 0$

$$x_1, x_2, x_3 = \prod_{i=1}^3 x_i$$

Likelihood function is the product of probability density function or probability mass function.

Question: Let $x_1, x_2, x_3, \dots, x_m$ be a random sample from Binomial distribution with parameters θ .

Find maximum likelihood estimate of θ .

Solution: We know,

For binomial distribution,

$$f(x_i; \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad x = 0, 1, 2, \dots, n$$

Step-1: Likelihood function is

$$L(\theta) = \prod_{i=1}^m f(x_i; \theta)$$

$$= \prod_{i=1}^m \binom{n}{x_i} \theta^{x_i} (1-\theta)^{n-x_i}$$

$$= \binom{n}{x_1} \theta^{x_1} (1-\theta)^{n-x_1} \cdot \binom{n}{x_2} \theta^{x_2} (1-\theta)^{n-x_2} \dots \binom{n}{x_m} \theta^{x_m} (1-\theta)^{n-x_m}$$

$$= [\theta^{x_1} \theta^{x_2} \dots \theta^{x_m}] [(1-\theta)^{n-x_1} (1-\theta)^{n-x_2} \dots (1-\theta)^{n-x_m}] \binom{n}{x_1} \binom{n}{x_2} \dots \binom{n}{x_m}$$

$$\begin{aligned}
&= \left[\theta^{x_1 + x_2 + \dots + x_m} \right] \left[(1-\theta)^{n-x_1 + n-x_2 + \dots + n-x_m} \right] \prod_{i=1}^m \binom{n}{x_i} \\
&= \left[\theta^{\sum_{i=1}^m x_i} \right] \left[(1-\theta)^{n+n+\dots+n-x_1-x_2-\dots-x_m} \right] \prod_{i=1}^m \binom{n}{x_i} \\
&= \left[\theta^{\sum_{i=1}^m x_i} \right] \left[(1-\theta)^{nm - (x_1 + x_2 + \dots + x_m)} \right] \prod_{i=1}^m \binom{n}{x_i} \\
&= \left[\theta^{\sum_{i=1}^m x_i} \right] \left[(1-\theta)^{nm - \sum_{i=1}^m x_i} \right] \prod_{i=1}^m \binom{n}{x_i}
\end{aligned}$$

Step-2:

$$\begin{aligned}
&\ln L(\theta) \\
&= \ln \left[\theta^{\sum x_i} (1-\theta)^{nm - \sum x_i} \prod_{i=1}^m \binom{n}{x_i} \right] \\
&= \ln \theta^{\sum x_i} + \ln (1-\theta)^{nm - \sum x_i} + \ln \left[\prod_{i=1}^m \binom{n}{x_i} \right] \quad [\because \ln(abc) = \ln a + \ln b + \ln c] \\
&= \sum x_i \ln \theta + (nm - \sum x_i) \ln(1-\theta) + \ln \left[\prod_{i=1}^m \binom{n}{x_i} \right] \quad [\because \ln a^n = n \ln a]
\end{aligned}$$

Step-3:

$$\begin{aligned}
&\frac{\partial}{\partial \theta} \ln L(\theta) \\
&= \frac{\partial}{\partial \theta} \left[\sum x_i \ln \theta + (nm - \sum x_i) \ln(1-\theta) + \ln \left[\prod_{i=1}^m \binom{n}{x_i} \right] \right] \\
&= \frac{\sum x_i}{\theta} - \frac{(nm - \sum x_i)}{1-\theta}
\end{aligned}$$

$$\boxed{
\begin{aligned}
\frac{\partial}{\partial \theta} \ln(1-\theta) &= \frac{1}{1-\theta} \cdot (-1) \\
&= -\frac{1}{1-\theta}
\end{aligned}
}$$

Step - 4:

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$\Rightarrow \frac{\sum x_i}{\theta} - \frac{(nm - \sum x_i)}{1-\theta} = 0$$

$$\Rightarrow \frac{(1-\theta) \sum x_i - \theta (nm - \sum x_i)}{\theta(1-\theta)} = 0$$

$$\Rightarrow \frac{\sum x_i - \cancel{\theta \sum x_i} - nm\theta + \cancel{\theta \sum x_i}}{\theta(1-\theta)} = 0$$

$$\Rightarrow \sum x_i - nm\theta = 0$$

$$\Rightarrow nm\theta = \sum x_i$$

$$\Rightarrow \theta = \frac{\sum_{i=1}^m x_i}{nm}$$

$$\boxed{\frac{\sum_{i=1}^m x_i}{m} = \frac{x_1 + x_2 + \dots + x_m}{m} = \hat{x}}$$

$$\Rightarrow \theta = \frac{\hat{x}}{n}$$

$\therefore$ This is the maximum likelihood of θ .

Question: Let $x_1, x_2, x_3, \dots, x_m$ be a random sample from poisson distribution with parameter λ . Find maximum likelihood estimate of λ .

Solution: We know,

For Poisson distribution, $f(x_i; \lambda) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$ $x = 0, 1, 2, \dots$

Step-1: Likelihood function is

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^m f(x_i; \lambda) \\ &= \prod_{i=1}^m \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \\ &= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \cdot \dots \cdot \frac{e^{-\lambda} \lambda^{x_m}}{x_m!} \\ &= (e^{-\lambda} \cdot e^{-\lambda} \cdot \dots \cdot e^{-\lambda}) \cdot (\lambda^{x_1} \lambda^{x_2} \dots \lambda^{x_m}) \cdot \left(\frac{1}{x_1!} \cdot \frac{1}{x_2!} \cdot \dots \cdot \frac{1}{x_m!} \right) \\ &= (e^{-m\lambda}) \cdot (\lambda^{x_1 + x_2 + \dots + x_m}) \cdot \prod_{i=1}^m \frac{1}{x_i!} \\ &= (e^{-m\lambda}) \cdot (\lambda^{\sum_{i=1}^m x_i}) \cdot \prod_{i=1}^m \frac{1}{x_i!} \end{aligned}$$

Step-2: $\ln L(\lambda) = \ln \left[e^{-m\lambda} \cdot \lambda^{\sum_{i=1}^m x_i} \cdot \prod_{i=1}^m \frac{1}{x_i!} \right]$

$$\begin{aligned} &= \ln(e^{-m\lambda}) + \ln(\lambda^{\sum_{i=1}^m x_i}) + \ln\left(\prod_{i=1}^m \frac{1}{x_i!}\right) \\ &= -m\lambda \ln e + \sum_{i=1}^m x_i \ln \lambda + \ln\left(\prod_{i=1}^m \frac{1}{x_i!}\right) \\ &= -m\lambda + \sum_{i=1}^m x_i \ln \lambda + \ln\left(\prod_{i=1}^m \frac{1}{x_i!}\right) \end{aligned}$$

Step-3:

$$\begin{aligned}\frac{\partial}{\partial \lambda} \ln L(\lambda) &= \frac{\partial}{\partial \lambda} \left[-m\lambda + \sum x_i \ln \lambda + \ln \left(\prod_{i=1}^m \frac{1}{x_i!} \right) \right] \\ &= -m + \frac{\sum x_i}{\lambda}\end{aligned}$$

Step-4:

$$\begin{aligned}\frac{\partial}{\partial \lambda} \ln L(\lambda) &= 0 \\ \Rightarrow -m + \frac{\sum x_i}{\lambda} &= 0 \\ \Rightarrow \frac{-m\lambda + \sum x_i}{\lambda} &= 0 \\ \Rightarrow -m\lambda + \sum x_i &= 0 \\ \Rightarrow m\lambda &= \sum x_i \\ \Rightarrow \lambda &= \frac{\sum x_i}{m} \\ \Rightarrow \lambda &= \hat{x}\end{aligned}$$

This is the maximum likelihood estimate of λ .

Q Hypothesis Testing:

→ Hypothesis is an assumption about a parameter. This assumption may or may not be true. Hypothesis testing refers to the formal procedure used by the statisticians to accept or reject this hypothesis.

→ There are two types of statistical hypothesis

1. Null Hypothesis (H_0)

2. Alternative Hypothesis (H_1)

[An alternative hypothesis is what the researcher wants to prove] A null hypothesis is the inverse of alternative hypothesis.

→ Hypothesis testing has 4 steps -

Step-1: Null hypothesis

Alternative hypothesis

Step-2: Test statistic: Test statistic will give a calculated value, which will be used to take decision whether we accept or reject H_0 .

Step-3: Rejection region: If calculated value falls in the rejection region, we reject H_0 (null hypothesis).

Step-4: Comment: (Since the calculated value falls in the rejection region, so we reject H_0 (null hypothesis) or since the calculated value does not fall in the rejection region, so we cannot reject H_0 (null hypothesis)).

→ Hypothesis test for the mean (μ)

Case 1: X has a normal distribution with known population variance (σ^2).

Case 2: X has a normal distribution with unknown population variance (σ^2).

Case 3: X has a general distribution, but we have a large sample size ($n \geq 30$). [Both known / unknown]

For Case 1, Case 2 and Case 3

$H_0: \mu = \mu_0$ → [Null hypothesis]

$H_1: \mu > \mu_0$

or, $H_1: \mu < \mu_0$ → [Alternative hypothesis]

or, $H_1: \mu \neq \mu_0$

For example, we want to test NSU student's average height is greater than 5ft.

$H_0: \mu = 5$

$H_1: \mu > 5$

If we want to test NSU student's average height is ^{less} than 5 ft.

$$H_0: \mu = 5$$

$$H_1: \mu < 5$$

If we want to test NSU student's average height is not equal to 5 ft.

$$H_0: \mu = 5$$

$$H_1: \mu \neq 5$$

Case 1:

$$\text{Test statistic is } \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0, 1) \quad \begin{matrix} \text{mean} \\ \text{variance} \end{matrix}$$

Here, $\bar{x}$ (Sample mean)

μ_0 (Given)

σ (Population standard deviation)

n (Sample size)

α	Z_α
0.10	1.282
0.05	1.645
0.025	1.960
0.01	2.326
0.005	2.576

When, $H_1: \mu > \mu_0$

The rejection region is $[Z_\alpha, +\infty[$

When, $H_1: \mu < \mu_0$

The rejection region is $]-\infty, -Z_\alpha]$

When $H_1: \mu \neq \mu_0$

The rejection region is $]-\infty, -Z_{\frac{\alpha}{2}}] \cup [Z_{\frac{\alpha}{2}}, +\infty[$

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression **that takes longer**. To test whether this impression is correct a sample ($n=12$) is taken with $\bar{x} = 92.2$. We assume that the production time is normal with $\sigma^2 = 144$. Verify whether this impression is correct at 5% level of significance.

Solution:

Null hypothesis, $H_0: \mu = 89$

Alternative hypothesis, $H_1: \mu > 89$

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0,1)$$

$$= \frac{92.2 - 89}{\sqrt{\frac{144}{12}}}$$

$$= 0.9237$$

$$\text{The rejection region} = [Z_{\alpha}, +\infty[$$

$$= [Z_{0.05}, +\infty[$$

$$= [1.645, +\infty[$$

Comment: Since test statistic's value (0.9237) does not fall in the rejection region, we can't reject H_0 at 5% level of significance. So, the factory owner's impression is incorrect.

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression that **takes lower**. To test whether this impression is correct a sample ($n=12$) is taken with $\bar{x} = 92.2$. We assume that the production time is normal with $\sigma^2 = 144$. Verify whether this impression is correct at 5% level of significance.

Solution:

$$H_0 : \mu = 89$$

$$H_1 : \mu < 89$$

$$\begin{aligned}\text{Test statistic} &= \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0,1) \\ &= \frac{92.2 - 89}{\sqrt{\frac{144}{12}}} \\ &= 0.9237\end{aligned}$$

$$\begin{aligned}\text{The rejection region} &=]-\infty, -Z_\alpha] \\ &=]-\infty, -Z_{0.05}] \\ &=]-\infty, -1.645]\end{aligned}$$

Comment: Since test statistics value (0.9237) does not fall in the rejection region, we cannot reject H_0 at 5% level of significance.

So, the factory owners impression is incorrect.

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression that **it does not take 89 min.** To test whether this impression is correct a sample ($n=12$) is taken with $\bar{x} = 92.2$. We assume that the production time is normal with $\sigma^2 = 144$. Verify whether this impression is correct at 5% level of significance.

Solution:

$$H_0: \mu = 89$$

$$H_1: \mu \neq 89$$

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0,1)$$

$$= \frac{92.2 - 89}{\sqrt{\frac{144}{12}}}$$

$$= 0.9237$$

$$\text{The rejection region} =]-\infty, -Z_{\frac{\alpha}{2}}] \cup [Z_{\frac{\alpha}{2}}, +\infty[$$

$$=]-\infty, -Z_{\frac{0.05}{2}}] \cup [Z_{\frac{0.05}{2}}, +\infty[$$

$$=]-\infty, -Z_{0.025}] \cup [Z_{0.025}, +\infty[$$

$$=]-\infty, -1.96] \cup [1.96, +\infty[$$

Comment: Since test statistic's value (0.9237) does not fall in the rejection region, so we can not reject H_0 at 5% level of significance.

So, the factory owners impression is incorrect.

Case 2:

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{(n-1)}$$

Here, s^2 indicates sample variance where

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \quad [\bar{x} = \text{sample mean}]$$

Example: 75 min, 78 min, 80 min, 92 min, 93 min

$$\bar{x} = \frac{75 + 78 + 80 + 92 + 93}{5} = 83.6$$

$$s^2 = \frac{(75-83.6)^2 + (78-83.6)^2 + (80-83.6)^2 + (92-83.6)^2 + (93-83.6)^2}{(5-1)}$$

$$= 69.3$$

when $H_1: \mu > \mu_0$

The rejection region is $[t_{\alpha, n-1}, +\infty[$

when $H_1: \mu < \mu_0$

The rejection region is $]-\infty, -t_{\alpha, n-1}]$

when $H_1: \mu \neq \mu_0$

The rejection region is $]-\infty, -t_{\frac{\alpha}{2}, n-1}] \cup [t_{\frac{\alpha}{2}, n-1}, +\infty[$

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression that it takes longer.

To test whether this impression is correct a sample ($n=5$) is taken

87 89 90 92 88

Verify this impression is correct at significance level 10%.

Solution:

$$H_0: \mu = 89$$

$$H_1: \mu > 89$$

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{(n-1)}$$

$$\text{Here, } \bar{x} = \frac{87 + 89 + 90 + 92 + 88}{5} = 89.2$$

$$\mu_0 = 89, n = 5$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(87-89.2)^2 + (89-89.2)^2 + (90-89.2)^2 + (92-89.2)^2 + (88-89.2)^2}{5-1}$$

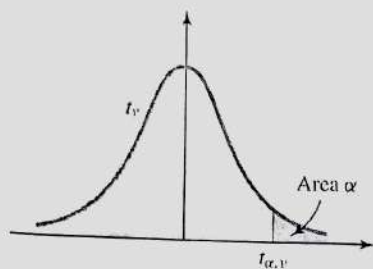
$$= 3.7$$

$$\therefore \text{Test statistic} = \frac{89.2 - 89}{\sqrt{\frac{3.7}{5}}} = 0.2325$$

$$\begin{aligned} \therefore \text{The rejection region} &= [t_{\alpha, (n-1)}, +\infty[\\ &= [t_{0.1, (5-1)}, +\infty[\\ &= [1.533, +\infty[\end{aligned}$$

Comment: Since the statistic value doesn't fall in the region, so we cannot reject H_0 (Null Hypothesis).

Table III: Critical Points of the t -Distribution



Degrees of freedom ν	α						
	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

$(n-1)$

α

t -distribution

$t_{(n-1)}$

degrees of freedom

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression that it takes lower.

To test whether this impression is correct a sample ($n=5$) is taken

87 89 90 92 88

Verify this impression is correct at significance level 10%.

Solution:

$$H_0: \mu = 89$$

$$H_1: \mu < 89$$

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{(n-1)}$$

$$\text{Hence, } \bar{x} = \frac{87 + 89 + 90 + 92 + 88}{5} = 89.2$$

$$\mu_0 = 89, \quad n = 5$$

$$\begin{aligned} s^2 &= \frac{\sum (x_i - \bar{x})^2}{n-1} \\ &= \frac{(87 - 89.2)^2 + (89 - 89.2)^2 + (90 - 89.2)^2 + (92 - 89.2)^2 + (88 - 89.2)^2}{5-1} \\ &= 3.7 \end{aligned}$$

$$\therefore \text{Test statistics} = \frac{89.2 - 89}{\sqrt{\frac{3.7}{5}}} = 0.2325$$

$$\begin{aligned} \therefore \text{The rejection region} &= [-\infty, -t_{\alpha, (n-1)}] \\ &= [-\infty, -t_{0.1, (5-1)}] \\ &= [-\infty, -1.533] \end{aligned}$$

Comment: Since the test statistics value doesn't fall in the rejection region, so we cannot reject H_0 (Null Hypothesis).

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression that **it does not takes 89 min.**

To test whether this impression is correct a sample ($n=5$) is taken

87 89 90 92 88

Verify this impression is correct at significance level 10%.

Solution:

$$H_0: \mu = 89$$

$$H_1: \mu \neq 89$$

$$\text{Test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{(n-1)}$$

$$\text{Hence, } \bar{x} = \frac{87 + 89 + 90 + 92 + 88}{5} = 89.2$$

$$\mu_0 = 89, n = 5$$

$$\begin{aligned} s^2 &= \frac{\sum (x_i - \bar{x})^2}{n-1} \\ &= \frac{(87-89.2)^2 + (89-89.2)^2 + (90-89.2)^2 + (92-89.2)^2 + (88-89.2)^2}{5-1} \\ &= 3.7 \end{aligned}$$

$$\text{Test statistic} = \frac{89.2 - 89}{\sqrt{\frac{3.7}{5}}} = 0.2325$$

$$\begin{aligned} \text{The rejection region} &=]-\infty, -t_{\frac{\alpha}{2}, (n-1)}] \cup [t_{\frac{\alpha}{2}, (n-1)}, +\infty[\\ &=]-\infty, -t_{\frac{0.1}{2}, (5-1)}] \cup [t_{\frac{0.1}{2}, (5-1)}, +\infty[\\ &=]-\infty, -t_{0.05, 4}] \cup [t_{0.05, 4}, +\infty[\\ &=]-\infty, -2.132] \cup [2.132, +\infty[\end{aligned}$$

Comment: Since the test statistic's value doesn't fall in the rejection region, so we cannot reject H_0 (Null Hypothesis).

Case 3: X has a general distribution, but we have a large sample size.

⊕ When population variance is known,

(σ^2 is known)

$$\text{The test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0,1)$$

⊕ When population variance is unknown,

(σ^2 is unknown)

$$\text{The test statistic} = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim N(0,1)$$

where, $\bar{x}$ = sample mean

s^2 = sample variance

n = sample size

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$H_1: \mu > \mu_0$$

The rejection region is $[Z_\alpha, +\infty[$

$$H_1: \mu < \mu_0$$

The rejection region is $]-\infty, -Z_\alpha]$

$$H_1: \mu \neq \mu_0$$

The rejection region is $]-\infty, -Z_{\frac{\alpha}{2}}] \cup [Z_{\frac{\alpha}{2}}, +\infty[$

Example: From a long-term experience, a factory owner knows that a worker can produce a product in an average time of 89 min. However, on Monday morning, there is the impression **that takes longer**. To test whether this impression is correct a sample ($n=50$) is taken with sample mean $\bar{x} = 92.2$ and sample variance $s^2 = 144$. Verify whether this impression is correct at 5% level of significance.

Solution:

$$H_0 : \mu = 89$$

$$H_1 : \mu > 89$$

$$\begin{aligned}\text{Test statistic} &= \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim N(0,1) \\ &= \frac{92.2 - 89}{\sqrt{\frac{144}{50}}} \\ &= 1.88\end{aligned}$$

$$\begin{aligned}\text{The rejection region} &= [Z_\alpha, +\infty[\\ &= [Z_{0.05}, +\infty[\\ &= [1.645, +\infty[\end{aligned}$$

Comment: Since test statistic's value (1.88) falls in the rejection region, so we reject H_0 at 5% level of significance.

That is the factory owners impression is correct.

④ Matched Pair t test: Matched - Pairs t - test is used to test whether there is a significant mean difference between two sets of paired data.

For example: Suppose we are interested in evaluating the effectiveness of a company training program. To evaluate this training effectiveness, measure the performance of the employees, before and after completing the program and analyze the differences by using matched pairs test.

Step 1: $H_0: \mu_D = 0$ $\rightarrow$ [Null hypothesis]

$H_1: \mu_D > 0$
or, $H_1: \mu_D < 0$
or, $H_1: \mu_D \neq 0$ $\rightarrow$ [Alternative hypothesis]

Here, $\mu_D = \mu_y - \mu_x$

Step 2: Test statistic = $\frac{\bar{D}}{\sqrt{\frac{s_D^2}{n}}} \sim t_{n-1}$

④ Here, D is the difference between paired value from two data sets.

$$\boxed{D = Y - X}$$

where, X is the variable in the first data set.

Y is the variable in the second data set.

$\boxed{\bar{D}}$ is the sample mean difference between paired observation.

$\boxed{s_D^2}$ is the variance of the differences.

$\boxed{n}$ is the number of paired observation.

Step 3: Rejection region,

$$H_1: \mu_D > 0$$

The rejection region is $[t_{\alpha, n-1}, +\infty[$

$$H_1: \mu_D < 0$$

The rejection region is $]-\infty, -t_{\alpha, n-1}]$

$$H_1: \mu_D \neq 0$$

The rejection region is $]-\infty, -t_{\alpha, n-1}] \cup [t_{\alpha, n-1}, +\infty[$

Step 4: Comment.

Example:

In 10 women the systolic blood pressure (mm Hg) is measured at the beginning of a clinical trial. Afterwards they have a fertility treatment with hormones. During this treatment they are again measured.

Id	before	during	Id	before	during
1	115	128	6	138	145
2	112	115	7	126	132
3	107	106	8	105	109
4	119	128	9	104	102
5	115	122	10	115	117

- Q: (a) Does the fertility treatment with hormones have a significant effect on the systolic blood pressure. Test this at 5% level of significance.
- (b) Does the fertility treatment with hormones increase the systolic blood pressure? Test this at 5% level of significance.
- (c) Does the fertility treatment with hormones decrease the systolic blood pressure? Test this at 5% level of significance.

Solution:

(a)

$$H_0 : \mu_D = 0$$

$$H_1 : \mu_D \neq 0$$

$$\text{Where, } \mu_D = \mu_y - \mu_x$$

μ_x = average blood pressure of population before fertility treatment with hormones.

μ_y = average blood pressure of population after fertility treatment with hormones.

$$\text{Test statistic} = \frac{\bar{D}}{\sqrt{\frac{s_D^2}{n}}} \sim t_{n-1}$$

From the data, we get

Id (i)	$D_i = Y_i - X_i$	(after - before)
1	13	$(128 - 115) = 13$
2	3	$(115 - 112) = 3$
3	-1	$(106 - 107) = -1$
4	9	$(128 - 119) = 9$
5	7	$(122 - 115) = 7$
6	7	$(145 - 138) = 7$
7	6	$(132 - 126) = 6$
8	4	$(109 - 105) = 4$
9	-2	$(102 - 104) = -2$
10	2	$(117 - 115) = 2$

$$\therefore \bar{D} = \frac{13 + 3 + (-1) + 9 + 7 + 7 + 6 + 4 + (-2) + 2}{10}$$

$$= 4.8$$

$$\begin{aligned}
 S_D^2 &= \frac{\sum_{i=1}^{10} (D_i - \bar{D})^2}{n-1} \\
 &= \frac{(D_1 - \bar{D})^2 + (D_2 - \bar{D})^2 + \dots + (D_{10} - \bar{D})^2}{10-1} \\
 &= \frac{(13-4.8)^2 + (3-4.8)^2 + \dots + (2-4.8)^2}{9} \\
 &= 20.844
 \end{aligned}$$

$$\begin{aligned}
 \text{Test statistic} &= \frac{4.8}{\sqrt{\frac{20.844}{10}}} \\
 &= 3.3247
 \end{aligned}$$

$$\begin{aligned}
 \text{Rejection region} &=]-\infty, -t_{\frac{\alpha}{2}, n-1}] \cup [t_{\frac{\alpha}{2}, n-1}, +\infty[\\
 &=]-\infty, -t_{\frac{0.05}{2}, 10-1}] \cup [t_{\frac{0.05}{2}, 10-1}, +\infty[\\
 &=]-\infty, -t_{0.025, 9}] \cup [t_{0.025, 9}, +\infty[\\
 &=]-\infty, -2.262] \cup [2.262, +\infty[
 \end{aligned}$$

Comment: Since the test statistics value falls in the rejection region, so we reject null hypothesis, i.e. The fertility treatment with hormones have a significant effect on the systolic blood pressure.

(b)

$$H_0: \mu_D = 0$$

$$H_1: \mu_D > 0$$

$$\text{Where, } \mu_D = \mu_y - \mu_x$$

μ_x = average blood pressure of population before fertility treatment with hormones.

μ_y = average blood pressure of population after fertility treatment with hormones.

$$\text{Test statistic} = \frac{\bar{D}}{\sqrt{\frac{s_D^2}{n}}} \sim t_{n-1}$$

From the data, we get

Id (i)	$D_i = Y_i - X_i$
1	13
2	3
3	-1
4	9
5	7
6	7
7	6
8	4
9	-2
10	2

$$\therefore \bar{D} = \frac{13+3+\dots+2}{10}$$

$$= 4.8$$

$$\begin{aligned} s_D^2 &= \frac{\sum_{i=1}^{10} (D_i - \bar{D})^2}{n-1} \\ &= \frac{(D_1 - \bar{D})^2 + (D_2 - \bar{D})^2 + \dots + (D_{10} - \bar{D})^2}{10-1} \\ &= \frac{(13-4.8)^2 + (3-4.8)^2 + \dots + (2-4.8)^2}{9} \\ &= 20.844 \end{aligned}$$

$$\therefore \text{Test statistic} = \frac{4.8}{\sqrt{\frac{20.844}{10}}} = 3.3247$$

$$\begin{aligned} \text{Rejection region} &= [t_{\alpha, n-1}, +\infty[\\ &= [t_{0.05, 9}, +\infty[\\ &= [1.833, +\infty[\end{aligned}$$

Comment: Since the test statistics values falls in the region, so we reject null hypothesis. i.e The fertility treatment with hormones increase the systolic blood pressure.

(c)

$$H_0 : \mu_D = 0$$

$$H_1 : \mu_D < 0$$

where, $\mu_D = \mu_Y - \mu_X$

μ_X = average blood pressure of population before fertility treatment with hormones.

μ_Y = average blood pressure of population after fertility treatment with hormones.

$$\text{Test statistic} = \frac{\bar{D}}{\sqrt{\frac{s_D^2}{n}}} \sim t_{n-1}$$

From the data, we get

Id (i)	$D_i = Y_i - X_i$
1	13
2	3
3	-1
4	9
5	7
6	7
7	6
8	4
9	-2
10	2

$$\therefore \bar{D} = \frac{13+3+\dots+2}{10} = 4.8$$

$$s_D^2 = \frac{\sum_{i=1}^{10} (D_i - \bar{D})^2}{n-1}$$

$$= \frac{(D_1 - \bar{D})^2 + (D_2 - \bar{D})^2 + \dots + (D_{10} - \bar{D})^2}{10-1}$$

$$= \frac{(13-4.8)^2 + (3-4.8)^2 + \dots + (-2-4.8)^2}{9}$$

$$= 20.844$$

$$\text{Test statistic} = \frac{4.8}{\sqrt{\frac{20.844}{10}}}$$

$$= 3.3247$$

$$\text{Rejection region} =]-\infty, -t_{\alpha, n-1}]$$

$$=]-\infty, t_{0.05, 9}]$$

$$=]-\infty, -1.833]$$

Comment: Since the test statistics value doesn't fall in the rejection region, so we can't reject null hypothesis. i.e. the fertility treatment with hormones does not decrease the systolic blood pressure.

④ Independent sample t test: also called the unpaired sample t test helps us to compare the means of two sets of data.

Example: In an experiment, we compare the result of treatment A and treatment B by seeing the survival time of mice.

Treatment A: 17 19 15 18 21 18

Treatment B: 18 15 13 16 13

④ Step-1: $H_0: \mu_1 = \mu_2 \longrightarrow H_0: \mu_1 - \mu_2 = 0$

$H_1: \mu_1 < \mu_2 \longrightarrow H_1: \mu_1 - \mu_2 < 0$

or $H_1: \mu_1 > \mu_2 \longrightarrow H_1: \mu_1 - \mu_2 > 0$

or $H_1: \mu_1 \neq \mu_2 \longrightarrow H_1: \mu_1 - \mu_2 \neq 0$

④ Step-2:

$$\text{Test statistic} = \frac{\bar{x} - \bar{y}}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

where,

$$s_p^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{n_1 + n_2 - 2}$$

Step-3:

If $H_1: \mu_1 - \mu_2 < 0$

The rejection region: $]-\infty, -t_{\alpha, n_1+n_2-2}]$

If $H_1: \mu_1 - \mu_2 > 0$

The rejection region: $[t_{\alpha, n_1+n_2-2}, +\infty[$

If $H_1: \mu_1 - \mu_2 \neq 0$

The rejection region: $]-\infty, -t_{\frac{\alpha}{2}, n_1+n_2-2}] \cup [t_{\frac{\alpha}{2}, n_1+n_2-2}, +\infty[$

Step-4: Comment.

Q Question: In an experiment, we compare the result of treatment A and treatment B by seeing the survival times of mice.

Treatment A: 17 19 15 18 21 18

Treatment B: 18 15 18 16 13

- Investigate whether treatment A gives better result? Test this at 5% level of significance.
- Investigate whether treatment B gives better result? Test this at 5% level of significance.
- Investigate whether treatment A and treatment B gives different results? Test this at 5% level of significance.

Solution:

(a) Null hypothesis, $H_0: \mu_1 = \mu_2$

$$\Rightarrow H_0: \mu_1 - \mu_2 = 0$$

Alternative hypothesis, $H_1: \mu_1 > \mu_2$

$$\Rightarrow H_1: \mu_1 - \mu_2 > 0$$

$$\therefore \text{Test statistic} = \frac{\bar{x} - \bar{y}}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

$$\text{Hence, } s_p^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{n_1 + n_2 - 2}$$

$$n_1 = 6$$

$$n_2 = 5$$

$$\text{Now, } \bar{x} = \frac{17 + 19 + 15 + 18 + 21 + 18}{6} = 18$$

$$\bar{y} = \frac{18 + 15 + 13 + 16 + 13}{5} = 15$$

$$\therefore s_p^2 = \frac{\sum_{i=1}^6 (x_i - \bar{x})^2 + \sum_{j=1}^5 (y_j - \bar{y})^2}{6 + 5 - 2}$$

$$= \frac{(17-18)^2 + \dots + (18-18)^2 + (18-15)^2 + \dots + (13-15)^2}{9}$$

$$= 4.22$$

$$\therefore \text{Test statistic} = \frac{18 - 15}{\sqrt{(4.22) \left(\frac{1}{6} + \frac{1}{5} \right)}}$$

$$= 2.41$$

$$\begin{aligned}\text{Rejection region: } & [t_{\alpha, n_1+n_2-2}, +\infty[\\ & = [t_{0.05, 9}, +\infty[\\ & = [1.833, +\infty[\end{aligned}$$

Comment: Since the calculated values fall in the rejection region, we reject H_0 . That is, treatment A gives better result.

$$\begin{aligned}\text{(b) Null hypothesis, } & H_0: \mu_1 = \mu_2 \\ & \Rightarrow H_0: \mu_1 - \mu_2 = 0\end{aligned}$$

$$\begin{aligned}\text{Alternative hypothesis, } & H_1: \mu_1 < \mu_2 \\ & \Rightarrow H_1: \mu_1 - \mu_2 < 0\end{aligned}$$

$$\text{Test statistic} = \frac{\bar{x} - \bar{y}}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1+n_2-2}$$

$$\text{Here, } s_p^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{n_1 + n_2 - 2}$$

$$\text{Now, } \bar{x} = \frac{17 + 19 + \dots + 18}{6} = 18$$

$$\bar{y} = \frac{18 + 15 + \dots + 13}{5} = 15$$

$$\begin{aligned}\therefore S_p^2 &= \frac{\sum_{i=1}^6 (x_i - \bar{x})^2 + \sum_{j=1}^5 (y_j - \bar{y})^2}{6+5-2} \\ &= \frac{(17-18)^2 + \dots + (18-18)^2 + (18-15)^2 + \dots + (13-15)^2}{9} \\ &= 4.22\end{aligned}$$

$$\begin{aligned}\therefore \text{Test statistic} &= \frac{18-15}{\sqrt{(4.22)\left(\frac{1}{6} + \frac{1}{5}\right)}} \\ &= 2.41\end{aligned}$$

$$\begin{aligned}\text{Rejection region: } &]-\infty, -t_{\alpha, n_1+n_2-2}] \\ &=]-\infty, -t_{0.05, 9}] \\ &=]-\infty, -1.833]\end{aligned}$$

Comment: Since the calculated value doesn't fall in the rejection region, so we can not reject H_0 . That is treatment B does not give better result.

(C)

$$H_0: \mu_1 = \mu_2$$

$$\Rightarrow \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2$$

$$\Rightarrow \mu_1 - \mu_2 \neq 0$$

$$\text{Test statistic} = \frac{\bar{x} - \bar{y}}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

Hence,

$$s_p^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{n_1 + n_2 - 2}$$

Now,

$$\bar{x} = \frac{17 + \dots + 18}{6} = 18$$

$$\bar{y} = \frac{18 + \dots + 13}{5} = 15$$

$$\therefore s_p^2 = \frac{\sum_{i=1}^6 (x_i - \bar{x})^2 + \sum_{j=1}^5 (y_j - \bar{y})^2}{6 + 5 - 2}$$

$$= \frac{(17-18)^2 + \dots + (18-18)^2 + (18-18)^2 + \dots + (13-18)^2}{9}$$

$$= 4.22$$

$$\therefore \text{Test statistic} = \frac{18 - 15}{\sqrt{(4.22) \left(\frac{1}{6} + \frac{1}{5} \right)}}$$

$$= 2.41$$

Rejection region: $]-\infty, -t_{\frac{\alpha}{2}, n_1+n_2-2}] \cup [t_{\frac{\alpha}{2}, n_1+n_2-2}, +\infty[$

$$=]-\infty, -t_{0.025, 9}] \cup [t_{0.025, 9}, +\infty[$$

$$=]-\infty, -2.262] \cup [2.262, +\infty[$$

Comment: Since the calculated value falls in the rejection region, so we reject H_0 , That is treatment A and treatment B give different results.

Quiz-1

1 Write down the sample space if you are tossing a coin and rolling a dice.

what is the probability of rolling an even numbers and flipping a tail?

what is the probability of rolling an even numbers given that head has flipped?

Solution:

Sample space of tossing a coin and rolling a dice
 $S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

Probability of rolling an odd numbers and flipping a tail is

$$P(\text{Odd \& T}) = \frac{3}{12} \quad \left[\begin{array}{l} \text{Odd \& Tail} = \{T1, T3, T5\} \\ \text{Total possible outcome} = 12 \end{array} \right]$$

Probability of rolling an even numbers given that head has flipped is calculated below.

Let, A be the event that even numbers is rolled.

B be the event that head has flipped.

$$\begin{array}{l|l|l} \therefore A \cap B = \{H2, H4, H6\} & B = \{H1, H2, H3, H4, H5, H6\} & \therefore P(A|B) = P(A \cap B) / P(B) \\ \therefore P(A \cap B) = \frac{3}{12} & \therefore P(B) = \frac{6}{12} & = \frac{3}{12} \times \frac{12}{6} \\ & & = \frac{1}{2} \\ & & \text{(Ans.)} \end{array}$$

- ② If you choose a card from a pack of cards,
what is the probability of a jack and a queen card?
what is the probability of a jack or a queen card?

Solution:

Probability of a jack and a queen card
is,

$$P(\text{Jack \& Queen}) = 0$$

[We can't choose both
jack and queen card
at once.]

Probability of a jack or a queen card is

$$P(\text{Jack or Queen}) = \frac{8}{52}$$

[4 Jack cards
4 Queen cards
Total possible outcomes = 52]

Assessment-1

The joint probability density function of X & Y are given below:

$$f(x, y) = \frac{2}{3} x^2 e^{-xy} \quad 1 < x < a \quad 0 < y < \infty$$

[1] Calculate the value of 'a'.

[2] Calculate the probability that X is less than 1.75 and Y is greater than 2.

Solution: [1] Here,

$$\int_1^a \int_0^{\infty} \frac{2}{3} x^2 e^{-xy} dy dx = 1$$

$$\Rightarrow \frac{2}{3} \int_1^a \int_0^{\infty} (e^{-xy}) dy x^2 dx = 1$$

$$\Rightarrow \frac{2}{3} \int_1^a \left[-\frac{e^{-xy}}{x} \right]_0^{\infty} x^2 dx = 1$$

$$\Rightarrow \frac{2}{3} \int_1^a \frac{1}{x} [-e^{-xy}]_0^{\infty} x^2 dx = 1$$

$$\Rightarrow \frac{2}{3} \int_1^a [(-0) - (-1)] \frac{1}{x} \cdot x^2 dx = 1$$

$$\Rightarrow \frac{2}{3} \int_1^a x dx = 1$$

$$\Rightarrow \frac{2}{3} \left[\frac{x^2}{2} \right]_1^a = 1$$

$$\Rightarrow \frac{2}{6} [a^2 - 1] = 1$$

$$\Rightarrow a^2 - 1 = \frac{6}{2}$$

$$\Rightarrow a^2 = 3 + 1$$

$$\Rightarrow a = \pm \sqrt{4}$$

$$\therefore a = -2, 2$$

(Ans!)

$$e^{-\infty} = 0$$

$$e^{\infty} = \infty$$

$$e^0 = 1$$

$$e^{-0} = 1$$

$$\boxed{2} \int_1^{\infty} \int_2^{1.75} \frac{2}{3} x^2 e^{-xy} dy dx$$

$$= \frac{2}{3} \int_1^{\infty} \int_2^{1.75} e^{-xy} dy \cdot x^2 dx$$

$$= \frac{2}{3} \int_1^{1.75} [-e^{-xy}]_2^{\infty} \frac{1}{x} \cdot x^2 dx$$

$$= \frac{2}{3} \int_1^{1.75} [(-0) - (-e^{-2x})] x dx$$

$$= \frac{2}{3} \int_1^{1.75} x e^{-2x} dx$$

$$= \frac{2}{3} \left[\left[-\frac{x e^{-2x}}{2} \right]_1^{1.75} + \int_1^{1.75} \frac{e^{-2x}}{2} dx \right]$$

$$= \frac{2}{3} \left[(-0.0264) - (-0.0677) + \left[-\frac{e^{-2x}}{4} \right]_1^{1.75} \right]$$

$$= \frac{2}{3} \left[0.0413 + [(-7.55 \times 10^{-3}) - (-0.0338)] \right]$$

$$= \frac{2}{3} [0.0413 + 0.02625]$$

$$= \frac{2}{3} \times (0.06755)$$

$$= 0.045$$

(Ans.)

$$\begin{aligned} u &= x \\ \Rightarrow \frac{d}{dx}(u) &= \frac{d}{dx}(x) \\ \Rightarrow \frac{du}{dx} &= 1 \\ \Rightarrow du &= dx \end{aligned}$$

$$\begin{aligned} u &= x \\ \Rightarrow du &= dx \end{aligned}$$

$$\int dv = \int e^{-2x} dx$$

$$\Rightarrow v = -\frac{e^{-2x}}{2}$$

* Parameter: (Is an unknown quantity of a distribution).

A parameter is a numerical summary of a population.

↓
[Group of individuals we are interested in.]

↓
[a number!]

Ex:

Population of interest

Parameters

[Adults in USA]

→ % that are married
→ average age
→ how many are older than 70 years old.

[Schools in Japan]

→ average number of students.
→ population of this school.

[# of red cars in US]

⇒ parameter!

Unknown! →

[Parameters are usually unknown]

[all cars in USA]

⇒ the population